

Research Implementation

Reliability-Aware Rank Aggregation for Deployment-Oriented Predictive Maintenance Model Selection

1. Introduction

Predictive maintenance increasingly uses machine-learning models to support industrial failure prediction, machinery-health monitoring and Remaining Useful Life (RUL) estimation [1]-[5]. However, practical model deployment involves more than selecting the model with the highest predictive accuracy. A candidate may perform strongly according to predictive quality while being less suitable according to robustness, probability reliability, computational efficiency or operational consequences [6], [10]-[12]. This creates a deployment-oriented model-selection problem in which several valid criteria can favour different candidate configurations.

Multi-criteria decision-making, algorithm-selection and rank-aggregation research provides established mechanisms for combining competing criteria or rankings [12]-[16]. Nevertheless, conventional equal aggregation assumes that criterion-specific rankings should exert the same influence on the final consensus. This assumption may be problematic when repeated experimental evidence shows that some criterion rankings are substantially more reproducible than others.

The implementation developed in this research addresses this issue through **Reliability-Aware Deployment Rank Aggregation (RDRA)**. RDRA estimates how reproducibly each deployment criterion ranks the candidate configurations across repeated experiments and uses this empirical reproducibility as reliability information during consensus aggregation.

Accordingly, the implementation addresses the following Main Research Question:

How can reliability-aware rank aggregation produce a robust and defensible consensus ranking of predictive-maintenance model configurations when deployment criteria provide conflicting and differently stable empirical evidence?

The proposed method has been implemented as a functional Python/Google Colab computational research artefact. The implementation is not intended to be a production industrial application or graphical user interface. Instead, its purpose is to demonstrate that the conceptual solution can be implemented reproducibly, produce tangible deployment-ranking outputs, and be subjected to controlled scientific validation.

The primary implementation uses the AI4I 2020 Predictive Maintenance Dataset [6] because the experimental framework provides repeated criterion-level evidence required for empirical reliability estimation. NASA C-MAPSS FD001 [7] is used as a complementary RUL-prediction task to examine deployment-ranking responsiveness in a substantially different sequential

degradation setting [22], [23]. The implemented methodology deliberately treats AI4I as the full RDRA development and validation environment, whereas C-MAPSS supplies complementary cross-task evidence rather than a second complete reliability-weighted replication.

2. Identified Challenges and Motivation

2.1 Challenge 1: Conflicting Deployment Criteria

The first challenge is that predictive-maintenance model configurations can be evaluated differently depending on the deployment criterion.

The primary AI4I implementation operationalises five deployment dimensions:

Table 2.1 Five deployment dimensions

Deployment dimension	Experimental evidence	Deployment purpose
Predictive Quality	F1-score and PR-AUC	Measures predictive performance, particularly failure-class performance
Robustness	Composite corruption robustness derived from performance retention under controlled sensor corruption	Measures resilience when sensor evidence is degraded
Probability Reliability	Brier score and Expected Calibration Error	Measures trustworthiness of probabilistic predictions
Computational Efficiency	Inference latency and model size	Represents computational deployment requirements
Operational Impact	False-negative / false-positive operational cost	Represents asymmetric practical consequences

These five dimensions are the deployment dimensions operationalised within the present study and are not intended to represent an exhaustive set of all possible industrial deployment considerations.

Calibration is treated separately from conventional discrimination because a model can produce strong classification performance while still producing poorly calibrated probabilities [24]. Similarly, operational error costs can make false negatives substantially more consequential than false positives in predictive-maintenance decision-making [12]. Therefore, the practical problem is that the best predictive model is not necessarily the best deployment model.

A configuration ranked first for Predictive Quality may not rank first for Efficiency or Operational Impact. Consequently, the implementation needs to preserve these separate perspectives and combine them only after criterion-specific evidence has been constructed.

Implementation justification

The raw performance measures are not immediately collapsed into a single overall score because doing so would obscure the criterion conflicts that motivate RDRA. Instead, the five deployment dimensions are preserved separately until criterion-specific rankings have been constructed, allowing disagreement between deployment perspectives to remain visible to the subsequent aggregation mechanism.

2.2 Challenge 2: Equal Aggregation Assumes Equal Evidence Confidence

A transparent baseline for combining criterion-specific rankings is Equal Borda aggregation. With five deployment dimensions, Equal Borda gives each criterion identical influence:

$$w_j^B = \frac{1}{5} = 0.20$$

Equal weighting is simple, interpretable and avoids introducing subjective priorities. However, it also makes an important assumption: the criterion-specific rankings are treated as though the empirical evidence supporting them warrants equal influence in the final consensus.

This assumption is not necessarily appropriate for the present research because the five deployment dimensions are derived from different forms of experimental evidence. Predictive Quality is based on predictive-performance measurements, Robustness on controlled sensor-corruption behaviour, Probability Reliability on calibration evidence, Computational Efficiency on latency and model-size measurements, and Operational Impact on asymmetric prediction-cost scenarios. Thus, there is no reason to assume in advance that these different evidence sources will reproduce the same candidate ordering with identical consistency across repeated experiments.

This creates the central challenge addressed by RDRA: If criterion-specific rankings are supported by different levels of empirical reproducibility, should they still contribute equally to the final deployment consensus?

RDRA addresses this challenge by estimating the reproducibility of each criterion-specific candidate ranking from repeated experimental evidence and using that reproducibility as empirical reliability information during aggregation. The purpose is not to replace multi-criteria decision-making with a new form of subjective weighting, but to introduce an evidence-confidence layer between experimental evaluation and consensus formation.

Therefore, a critical distinction emerges: empirical reliability weights reflect evidence reproducibility, whereas stakeholder-preference weights reflect decision priorities, and these two concepts should not be conflated.

A stakeholder-preference weight represents how important a deployment criterion is to a decision-maker. In contrast, an RDRA reliability weight represents how consistently repeated experimental evidence supports that criterion's candidate ordering. Weighted multi-criteria methods already support preference- or importance-based weighting [13], [14]. The proposed RDRA mechanism instead derives its weighting information from observed ranking reproducibility.

The repeated AI4I experiments subsequently provide the evidence required to estimate these criterion-specific reliability values. The detailed Leave-One-Repetition-Out reliability-estimation procedure, implementation code and empirical reliability-weight output are presented in Sections 5.3.3 and 5.3.4. Stakeholder preference remains conceptually separate and is evaluated later through VQ3 in Section 6.

2.3 Challenge 3: Stability Alone Can Produce Misleading Validation

A third challenge concerns how the quality of a deployment-ranking method should be evaluated. A ranking method should not be considered effective simply because its output changes very little across different experimental conditions.

For instance, a hypothetical aggregation method that always returns the same candidate ordering could appear perfectly stable. However, such behaviour would be undesirable if the underlying deployment evidence changed substantially and the ranking failed to respond. In that situation, the method would be stable but disconnected from the evidence it is intended to represent.

Therefore, the evaluation must clearly distinguish between two separate properties: Consistency and Responsiveness.

Consistency refers to the ability of the aggregation method to avoid unnecessary or disproportionate ranking changes when the available deployment criteria vary. A useful consensus should not change dramatically because of relatively limited structural uncertainty in the criterion set.

In contrast, responsiveness refers to the ability of the ranking to change when meaningful changes occur in the supporting deployment evidence, such as deterioration caused by sensor corruption.

These properties are complementary rather than contradictory. A defensible deployment-ranking method should resist unnecessary change while remaining capable of responding to meaningful evidence change.

Excessive instability would make a model recommendation difficult to defend, whereas excessive stability could make the ranking insensitive to genuine deterioration in the evidence supporting deployment suitability. Therefore, stability alone is not a sufficient validation criterion.

This challenge directly determines the scientific evaluation framework used in the research. Three complementary validation questions are employed:

- **VQ1 - Criterion Uncertainty:** evaluates whether the consensus remains reasonably consistent when the available deployment criteria change.
- **VQ2 - Data-Quality Uncertainty:** evaluates whether the ranking responds when meaningful changes occur in the underlying deployment evidence.
- **VQ3 - Preference Uncertainty:** evaluates whether the resulting ranked shortlist remains useful when stakeholder deployment priorities vary.

The three validation perspectives aim to address related but different questions. VQ1 examines resistance to unnecessary ranking disruption, VQ2 examines evidence-responsive behaviour, and VQ3 determines whether ranking-level behaviour translates into useful downstream model-selection choices.

This separation is important because better ranking consistency does not imply better downstream decision usefulness. Similarly, stronger responsiveness does not guarantee superior performance on every downstream decision metric.

Consequently, the proposed method is not evaluated under the assumption that RDRA must outperform Equal Borda on every measure. The validation framework is deliberately falsifiable: the additional reliability-aware mechanism should be considered justified only where it provides meaningful improvement on the properties it is designed to target without introducing unacceptable behaviour elsewhere.

Hence, the evaluation asks that under which uncertainty conditions does reliability-aware aggregation provide measurable value beyond simpler equal aggregation?

The detailed VQ1, VQ2 and VQ3 experimental designs, statistical tests and empirical findings are presented in **Section 6: Solution Evaluation and Scientific Testing**.

3. Proposed Conceptual Solution: RDRA

The conceptual solution proposed in this research is **Reliability-Aware Deployment Rank Aggregation (RDRA)**. RDRA is designed for deployment-oriented predictive-maintenance model selection where candidate configurations are evaluated according to multiple potentially conflicting criteria and where the empirical evidence supporting those criterion-specific rankings may differ in reproducibility.

The approach builds on established concepts in multi-criteria decision-making and rank aggregation [13]-[16], [20] but introduces an explicit empirical reliability-estimation layer before final consensus formation. Rather than if all criterion-specific rankings should contribute equally,

RDRA estimates how reproducibly each criterion orders the candidate configurations across repeated experimental evidence and uses that reproducibility to determine relative aggregation influence.

The resulting conceptual sequence begins with repeated experimental evidence, which is converted into criterion-specific rankings. These rankings are analysed for reproducibility to produce empirical reliability weights, which are then applied to normalised Borda utilities to generate the RDRA consensus ranking.

Thus, this method separates two questions that conventional equal aggregation treats identically: How does each deployment criterion rank the candidate models? and How reproducibly is that criterion-specific ranking supported by repeated evidence?

The conceptual method has subsequently been operationalised as a functioning Python/Google Colab computational research artefact. The detailed implementation is presented in Section 5, while this section defines the architecture, mathematical formulation, novelty and expected benefits of the proposed solution.

3.1 Conceptual Architecture

Figure 1 presents the overall implemented RDRA architecture and validation framework used in this research.

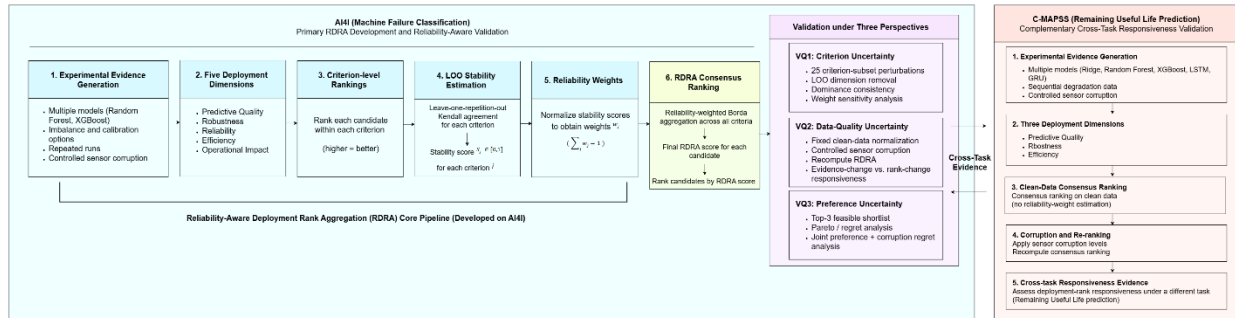


Figure 3.1 Implemented RDRA architecture and validation framework

The primary branch of the architecture uses repeated AI4I experimental evidence [6]. Candidate configurations are evaluated across the five deployment dimensions operationalised in this study: Predictive Quality, Robustness, Probability Reliability, Computational Efficiency and Operational Impact. These dimensions were defined in Section 2.1 and are retained separately so that each can produce its own candidate ordering before aggregation.

For each criterion, the implementation derives a candidate ranking and then estimates how reproducibly that ordering is supported across repeated experimental runs using Leave-One-Repetition-Out (LOO) Kendall rank agreement [19]. The resulting criterion-specific stability

estimates are converted into empirical reliability weights and supplied to a reliability-weighted Borda aggregation procedure. The output is a final deployment-oriented RDRA consensus ranking.

The architecture also separates method construction from method evaluation. VQ1, VQ2 and VQ3 are not part of the RDRA score itself. Instead, they test different properties of the implemented solution. VQ1 examines consistency under criterion uncertainty, VQ2 examines responsiveness to meaningful changes in deployment evidence, and VQ3 examines downstream shortlist usefulness under changing stakeholder preferences. Their detailed designs and results are presented in Section 6.

NASA C-MAPSS FD001 is incorporated as a separate complementary branch [22], [23]. It broadens the experimental context to sequential Remaining Useful Life prediction but is not treated as a second complete reliability-weighted replication because the current C-MAPSS workflow does not provide the same repeated five-dimension evidence structure used for full AI4I RDRA estimation.

Implementation Justification

The separation between predictive-model evaluation and rank aggregation is deliberate. RDRA operates as a deployment-selection layer rather than being embedded within the internal structure of Random Forest, XGBoost or any other predictive model. This makes the proposed mechanism comparatively modular: candidate-model families can change without altering the fundamental aggregation logic, provided appropriate criterion-level evidence is available.

Architecture also prevents the existence of empirical reliability weights from being treated as evidence of superiority by itself. Reliability estimation constructs the proposed method; the later validation framework determines whether the additional mechanism produces useful behaviour under the uncertainty conditions being studied.

3.2 Mathematical Formulation

The primary RDRA formulation converts repeated criterion-level evidence into a reliability-aware consensus ranking. Figure 2 summarises the internal mathematical sequence.

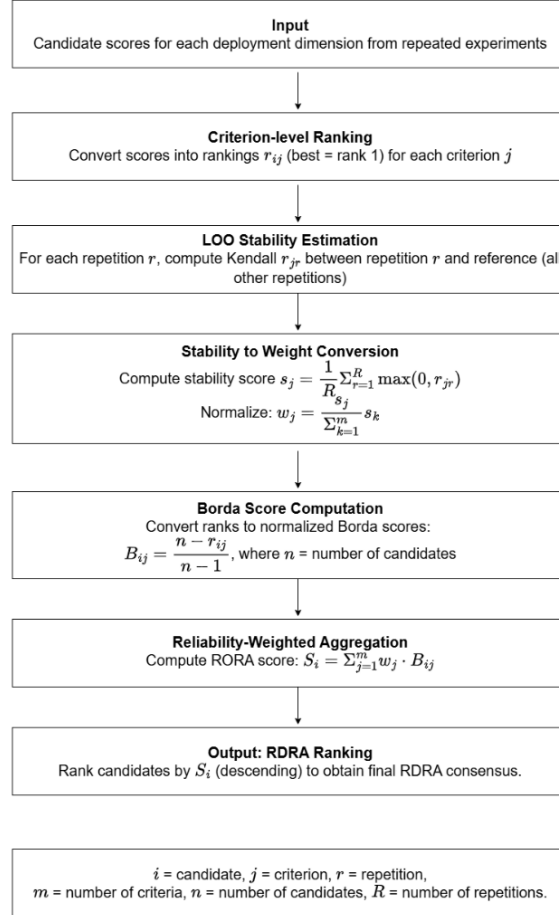


Figure 3.2 Algorithmic and mathematical workflow of primary RDRA

The notation used throughout the formulation is summarised below.

Table 3.1 Symbol definitions used in the RDRA formulation

Symbol	Meaning
i	Candidate predictive-maintenance model/configuration
j	Deployment criterion
r	Experimental repetition
n	Number of candidate configurations
m	Number of deployment criteria
R	Number of experimental repetitions
r_{ij}	Final rank of candidate i under criterion j
τ_{jr}	LOO Kendall agreement for criterion j in repetition r

s_j	Empirical ranking-stability score for criterion j
w_j	Normalised empirical reliability weight for criterion j
B_{ij}	Normalised Borda utility of candidate i under criterion j
S_i	Final RDRA consensus score for candidate i

The distinction between r_{ij} and τ_{jr} is crucial. The former represents the position of a candidate within one criterion ranking, whereas the latter represents the agreement between two complete candidate rankings.

3.2.1 Criterion-Level Ranking

After the heterogeneous raw measurements have been transformed into consistently oriented deployment-dimension scores, candidate i receives a criterion-specific rank:

$$r_{ij}.$$

The best candidate under criterion j receives:

$$r_{ij} = 1.$$

Because all dimension scores have already been converted to a benefit-oriented interpretation, a higher dimension score represents better deployment evidence, and the candidates are therefore ranked in descending order.

Where candidates are tied, average ranks are used. For instance, two candidates sharing the second and third positions each receive:

$$r = \frac{2 + 3}{2} = 2.5$$

Average tie handling avoids introducing ordering information that is not supported by the evidence. The implementation details are presented in Section 5.3.2.

3.2.2 Leave-One-Repetition-Out Ranking Agreement

The defining reliability component of RDRA is the estimation of criterion-level ranking reproducibility.

For each deployment criterion j and repetition r , one experimental repetition is treated as held-out evidence. Its candidate ranking is compared with a reference ranking derived from the remaining repetitions.

The resulting agreement is:

$$\tau_{jr} = \tau\left(r_j^{(r)}, r_j^{(-r)}\right),$$

where:

- $r_j^{(r)}$ is the candidate ranking from the held-out repetition; and
- $r_j^{(-r)}$ is the reference ranking obtained from the remaining repetitions.

The procedure is repeated for:

$$r = 1, \dots, R.$$

For the primary AI4I implementation:

$$R = 20.$$

LOO agreement is used because a ranking calculated from all repetitions can show the final ordering but cannot reveal whether that ordering is consistently reproduced across individual experimental runs. Kendall agreement is appropriate because the quantity being evaluated is **ordinal ranking reproducibility**, rather than numerical similarity between raw metric values.

The executable LOO procedure is presented in Section 5.3.3.

3.2.3 Empirical Stability Estimation

The repetition-level Kendall agreements are converted into a criterion-specific empirical stability estimate:

$$s_j = \frac{1}{R} \sum_{r=1}^R \max(0, \tau_{jr}).$$

Therefore:

$$0 \leq s_j \leq 1.$$

A larger s_j indicates that criterion j reproduces its candidate ordering more consistently across the repeated evidence.

Negative Kendall agreement is deliberately treated as zero positive reliability evidence:

$$\max(0, \tau_{jr}).$$

This choice preserves the intended interpretation of RDRA. Reliability represents positive confidence in the reproducibility of a criterion-specific ordering. A negative agreement value should therefore reduce confidence rather than reverse the semantic direction of that criterion. Allowing negative stability to become a negative aggregation weight would create criterion inversion rather than low reliability. The detailed implementation, signed-agreement output and positive-stability calculation are presented in Section 5.3.3.

3.2.4 Conversion from Stability to Reliability Weight

The criterion-specific stability values are normalised into aggregation weights:

$$w_j = \frac{s_j}{\sum_{k=1}^m s_k}.$$

The resulting weights satisfy:

$$w_j \geq 0$$

and:

$$\sum_{j=1}^m w_j = 1.$$

Criteria whose candidate rankings are more reproducible therefore receive proportionally greater aggregation influence under the primary RDRA formulation.

Importantly, w_j represents empirical ranking reliability, not stakeholder importance. The distinction between reliability and preference was established in Section 2.2. If no positive stability evidence is available across the criteria, the implementation falls back to equal weighting because there would be no empirical basis for unequal influence.

The empirical weight construction and resulting AI4I reliability estimates are presented in Section 5.3.4.

3.2.5 Normalised Borda Utility

Criterion-specific ranks are converted into normalised Borda utilities:

$$B_{ij} = \frac{n - r_{ij}}{n - 1}.$$

For the highest-ranked candidate:

$$r_{ij} = 1 \Rightarrow B_{ij} = 1,$$

whereas for the lowest-ranked candidate:

$$r_{ij} = n \Rightarrow B_{ij} = 0.$$

Intermediate candidate ranks are mapped proportionally between 0 and 1.

The Borda transformation provides a common ordinal aggregation scale across heterogeneous deployment criteria. Borda-style aggregation itself is established within rank-aggregation

research [16]. The proposed RDRA contribution lies in how criterion influence is determined before these utilities are combined.

3.2.6 Reliability-Weighted Aggregation

The final RDRA score for candidate i is:

$$S_i = \sum_{j=1}^m w_j B_{ij}.$$

Each criterion contributes:

$$w_j B_{ij}$$

to the final candidate score.

Candidates are ranked in descending order of S_i to obtain the final RDRA consensus.

The aggregation therefore combines two distinct forms of information:

$$B_{ij} = \text{candidate position under criterion } j,$$

and

$$w_j = \text{empirical reproducibility of criterion } j.$$

A lower-reliability criterion is not automatically removed. Instead, its influence is moderated proportionally, allowing RDRA to preserve multiple deployment perspectives while reflecting differences in the empirical support for their rankings.

3.2.7 Equal Borda as the Controlled Baseline

Equal Borda uses:

$$w_j^B = \frac{1}{m}.$$

With five deployment criteria:

$$w_j^B = 0.20.$$

Its consensus score is:

$$S_i^B = \sum_{j=1}^m \frac{1}{m} B_{ij}.$$

Equal Borda then uses the same criterion rankings and normalised Borda framework as RDRA but assumes equal criterion influence. This establishes a controlled baseline: Equal Borda enforces equal influence across all criteria, while RDRA replaces that assumption with influence derived from empirical reliability estimates. The shared computational implementation of the two approaches is demonstrated in Section 5.3.5.

3.2.8 Primary RDRA and Exploratory Reliability-Influence Refinement

The equations above define the **primary RDRA formulation**, which is equivalent to:

$$w_j^{(1)} = \frac{s_j}{\sum_k s_k}.$$

An exploratory extension generalises the reliability-weight relationship as:

$$w_j(\gamma) = \frac{s_j^\gamma}{\sum_k s_k^\gamma}.$$

Under this formulation:

$\gamma = 0$ corresponds to the Equal Borda limit,

$\gamma = 1$ corresponds to primary RDRA, and

$\gamma > 1$ increases the relative influence of differences in empirical ranking reproducibility.

This extension is deliberately separated from the definition of primary RDRA. Its purpose is to investigate whether the strength of reliability **influence** affects ranking behaviour without retrospectively redefining the original method.

The development-only search selected exploratory configuration and associated safeguards are presented in Section 5.3.6. The primary proposed method remains the reliability-proportional formulation defined by $\gamma = 1$.

3.2.9 Mathematical-to-Implementation Correspondence

The mathematical formulation corresponds directly to implemented procedures in the Google Colab research artefact.

Table 3.2 Mathematical components to implemented procedures in the RDRA framework

Mathematical component	Implemented procedure
Criterion rank r_{ij}	rank_from_score()
LOO agreement τ_{jr}	Kendall agreement within the LOO procedure
Criterion stability s_j	estimate_dimension_stability_loo()
Reliability weight w_j	Normalisation of positive stability values

Borda utility B_{ij}	normalized_borda_from_rank()
Equal Borda consensus	aggregate_borda(..., reliability_weights=None)
RDRA score S_i	aggregate_borda(..., reliability_weights=...)
Final consensus rank	Descending tie-aware ranking of aggregate scores

This demonstrates that the mathematical proposal has progressed into an executable research artefact. The detailed implementation code and outputs are presented in Section 5.3.

3.3 Novelty and Innovation

The proposed contribution does not claim novelty in the individual mathematical techniques used within RDRA. Borda aggregation, Kendall rank agreement, weighted multi-criteria decision-making and rank aggregation are established approaches [13]-[16].

The methodological contribution instead lies in how criterion influence is derived and incorporated into deployment-oriented predictive-maintenance model selection.

The central adaptation is to estimate criterion-level empirical ranking reproducibility from repeated experimental evidence and use that reproducibility as reliability information within deployment-oriented rank aggregation.

Conceptually:

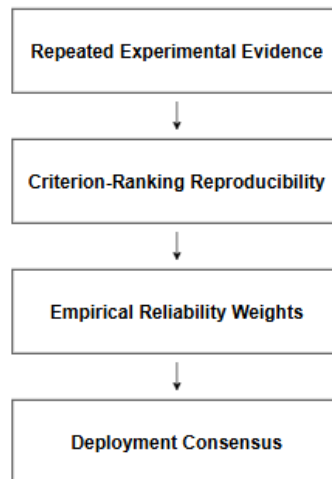


Figure 3.3 Sequential process from experimental evidence to deployment consensus

This introduces an explicit evidence-confidence layer between experimental evaluation and consensus formation.

Evidence-derived rather than preference-derived weighting

Weighted multi-criteria methods can assign unequal importance to criteria according to decision-maker priorities [13], [14]. RDRA uses unequal weights for a different reason: the weights represent how consistently repeated evidence supports a criterion-specific candidate ordering.

Thus, RDRA's reliability weight quantifies the empirical trustworthiness of the evidence, while stakeholder-preference weights encode the decision-maker's priorities. These two forms of influence are conceptually and methodologically distinct. This separation allows reliability and preference uncertainty to be studied independently.

Ranking reproducibility as the reliability quantity

RDRA defines reliability at the level of the object being aggregated: the candidate ranking.

This is different from relying only on variance in raw metrics. Two experimental repetitions can contain different absolute performance values while supporting nearly identical candidate orderings. Conversely, relatively small score changes may change the ranking when competing candidates are close.

Therefore, using LOO Kendall agreement connects empirical reliability directly to the ordinal information consumed by the aggregation procedure.

Deployment-oriented contribution

The method is designed for selection across multiple deployment concerns rather than predictive accuracy alone. The five operationalised dimensions represent predictive behaviour, robustness, calibration, computational requirements and operational consequences. Thus, RDRA provides a mechanism for forming a deployment consensus while retaining the distinct perspectives represented by these criteria.

Bounded novelty claim

The novelty claim is deliberately limited. This research does not claim that:

- Borda aggregation is new
- Kendall rank agreement is new
- weighted rank aggregation is new
- multi-criteria model selection is new
- reliability-aware aggregation must outperform equal aggregation in every deployment context.

The specific contribution investigated is: **a deployment-oriented rank-aggregation mechanism in which criterion influence is informed by empirical criterion-ranking reproducibility estimated from repeated predictive-maintenance experiments.**

This bounded position is crucial because methodological novelty does not itself demonstrate scientific superiority. The extent to which the proposed mechanism provides measurable value is tested separately in Section 6.

3.4 Expected Benefits

The conceptual design of RDRA provides several potential benefits for deployment-oriented predictive-maintenance model selection. These are treated as properties to be tested, rather than conclusions assumed from the mathematical formulation.

Evidence-aware aggregation

The principal expected benefit is that RDRA makes use of repetition-to-repetition ranking behaviour that would otherwise be discarded after average performance is calculated. Criteria supported by more reproducible candidate orderings can contribute proportionally more to consensus, while less reproducible criteria remain represented with reduced influence. This provides an evidence-aware alternative to automatically assuming equal confidence in every criterion ranking.

Traceability and interpretability

The RDRA consensus remains decomposable into criterion-specific ranks, Borda utilities, stability estimates and reliability weights. Consequently, a candidate's final position can be traced back through the evidence that produced it. This improves interpretability compared with a single opaque composite score and allows changes relative to Equal Borda to be explained in terms of the underlying criterion contributions.

Preservation of multiple deployment perspectives

RDRA maintains the deployment dimensions separately until the aggregation stage rather than replacing the deployment problem with a single predictive-performance measure. This preserves potentially important differences between predictive quality, robustness, probability reliability, efficiency and operational impact while allowing their evidence quality to influence the final consensus.

Modularity and model independence

The core aggregation procedure operates on criterion-level candidate evidence rather than on the internal architecture of a specific machine-learning model. This makes the approach conceptually applicable to different candidate-model families when appropriate deployment

evidence is available. Full reliability-aware weighting additionally requires repeated evidence from which ranking reproducibility can be estimated.

Separation of empirical reliability and stakeholder preference

RDRA enforces a conceptual separation between the empirical reliability of the evidence and the stakeholder's value-based preferences. The former quantifies how trustworthy the experimental evidence is, while the latter encodes the decision-maker's priorities. These two influences are methodologically distinct and should not be conflated.

The former informs RDRA aggregation, while the latter is examined separately through downstream preference analysis. This separation makes it possible to determine whether a criterion has greater influence because its evidence is more reproducible rather than because a stakeholder considers it more important.

Connection to scientific validation

The expected properties of RDRA are evaluated explicitly rather than assumed.

Table 3.3 Evaluation mechanisms for expected RDRA properties

Expected property	Evaluation mechanism
Alignment with deployment evidence	Dominance-consistency sanity check
Consistency under changing criteria	VQ1
Sensitivity to reliability-weight uncertainty	Reliability-weight perturbation analysis
Responsiveness to meaningful evidence change	VQ2
Downstream shortlist usefulness under changing preferences	VQ3
Breadth across a different predictive-maintenance task	Complementary C-MAPSS analysis

This mapping ensures that the conceptual advantages proposed in this section are subsequently subjected to empirical testing. Hence, section 6 determines which expected properties are supported under the implemented experimental conditions and where the evidence establishes a more limited contribution.

4. Scope, Requirements and Development Plan

The scope of the implemented solution is defined around the central research problem identified in Section 2: producing a deployment-oriented consensus ranking from multiple potentially conflicting criteria when the empirical evidence supporting those criterion-specific rankings differs in reproducibility.

The resulting artefact is a **computational research prototype for reliability-aware predictive-maintenance model selection**, rather than a complete industrial predictive-maintenance platform. Its purpose is to demonstrate that RDRA can be operationalised as a tangible solution, use experimentally generated evidence, produce traceable candidate rankings, and support controlled scientific evaluation.

The primary implementation is based on the AI4I 2020 Predictive Maintenance Dataset [6]. NASA C-MAPSS FD001 is incorporated separately to provide complementary evidence in a different Remaining Useful Life prediction setting [7], [22], [23]. The scope is deliberately constrained to functionality that directly supports development and validation of the proposed rank-aggregation mechanism rather than production features such as live sensor ingestion, maintenance scheduling or a graphical industrial interface.

4.1 Experimental and Implementation Scope

4.1.1 Primary AI4I Scope

AI4I 2020 [6] provides the primary environment for complete RDRA development and validation. The research considers a binary machine-failure classification task and evaluates combinations of:

- Random Forest
- XGBoost
- three imbalance-handling conditions
- three probability-calibration conditions.

This produces $2 \times 3 \times 3 = 18$ candidate model configurations.

Each configuration is evaluated across $R = 20$ experimental repetitions.

The repeated structure is essential to the scope of the primary implementation because empirical criterion-ranking reproducibility can only be estimated when repetition-level evidence is retained.

The AI4I branch provides evidence for all five deployment dimensions:

1. Predictive Quality
2. Robustness
3. Probability Reliability

4. Computational Efficiency
5. Operational Impact.

These five dimensions were selected because they represent distinct deployment concerns relevant to predictive-maintenance model selection. Predictive performance alone would not capture calibration reliability, sensitivity to degraded sensor evidence, deployment overhead or asymmetric operational consequences [1]–[6], [8], [12]. Thus, AI4I is used as the primary full RDRA environment because it provides both the five-dimensional deployment evidence and the repeated experimental structure required for empirical reliability estimation.

4.1.2 Complementary C-MAPSS Scope

NASA C-MAPSS FD001 [22], [23] extends the research to a sequential Remaining Useful Life prediction setting. The candidate model set includes:

- Ridge regression
- Random Forest
- XGBoost
- LSTM
- GRU

The current C-MAPSS implementation provides evidence for three deployment-oriented dimensions:

- Predictive Quality
- Robustness
- Computational Efficiency.

Controlled temporal sensor degradation is also incorporated to examine how deployment-oriented rankings respond when sequential data quality changes.

C-MAPSS is deliberately scoped as complementary cross-task deployment-rank responsiveness evidence rather than as a second complete RDRA validation.

The current C-MAPSS pipeline does not contain the same repeated five-dimension evidence structure required for an independent full reliability-weight estimation. This boundary allows the project to broaden its experimental context while keeping the strength of the resulting claim consistent with the implemented evidence.

4.2 Functional and Implementation Requirements

The requirements define the essential features and components that the research artefact must provide to operationalise RDRA and support its subsequent scientific evaluation. They cover the

complete pathway from experimental evidence preparation to consensus formation, traceability and controlled validation.

Table 4.1 Functional and implementation requirements of the RDRA artefact

ID	Functional requirement	Purpose
FR1	Load the generated AI4I experimental evidence	Provide the empirical input required by the RDRA pipeline
FR2	Construct five deployment-oriented dimensions from heterogeneous metrics	Represent predictive, robustness, reliability, efficiency and operational perspectives
FR3	Convert cost- and benefit-oriented evidence to a consistent preference direction	Ensure higher dimension scores consistently represent more desirable deployment evidence
FR4	Preserve repetition-level evidence	Retain the information required for empirical reliability estimation
FR5	Produce criterion-specific candidate rankings	Preserve the distinct ordering produced by each deployment dimension
FR6	Handle tied candidates consistently	Avoid introducing unsupported artificial ordering
FR7	Estimate criterion-ranking reproducibility using LOO agreement	Quantify empirical stability across repeated experiments
FR8	Convert empirical stability into normalised reliability weights	Produce the weighting information required by RDRA
FR9	Calculate normalised Borda utilities	Place criterion rankings on a common aggregation scale
FR10	Generate the primary RDRA consensus ranking	Implement the proposed conceptual solution
FR11	Generate an Equal Borda baseline	Provide a controlled equal-weight comparison
FR12	Retain the Existing Composite Method	Provide an additional simple aggregation reference
FR13	Verify basic evidence alignment through dominance consistency	Check that the consensus does not contradict straightforward dominance relationships
FR14	Evaluate criterion uncertainty	Implement VQ1
FR15	Evaluate reliability-weight sensitivity	Test whether the broader consensus depends excessively on one exact weight vector
FR16	Evaluate data-quality responsiveness	Implement VQ2
FR17	Evaluate downstream preference uncertainty	Implement VQ3
FR18	Export intermediate and final artefacts	Support traceability and reproducibility

FR19	Preserve separation between primary and exploratory RDRA	Maintain the distinction between primary $\gamma = 1$ RDRA and the locked exploratory refinement
FR20	Provide complementary C-MAPSS evidence	Extend evaluation to a different predictive-maintenance task

4.2.1 Requirement Rationale

The requirements follow the logical structure of the proposed solution. FR1-FR6 establish the criterion-level evidence required for deployment-oriented ranking, while FR7-FR10 implement the central reliability-aware aggregation mechanism. FR11-FR12 provide controlled comparison methods, and FR13-FR17 ensure that the artefact can be scientifically evaluated rather than judged only by whether it produces a final ranking. FR18-FR20 support traceability, methodological separation and complementary cross-task evidence.

Several requirements are important to the validity of the research. Repetition-level evidence must be retained until reliability estimation because criterion-ranking reproducibility cannot be recovered from a single averaged ranking. Equal Borda must also remain comparable with RDRA through the same underlying ranking framework so that the effect of reliability-informed criterion influence can be examined independently. The detailed implementation of these requirements is presented in Section 5, while their achievement is assessed explicitly in Section 6.

4.3 Current Development phase and status

At the time of this Research Implementation report, the principal components required for the proposed solution have reached broad completion. The project has progressed from conceptual formulation to a functioning computational artefact with implemented ranking, reliability-estimation, aggregation and validation components.

Table 4.2 Current RDRA development status

Development component	Current status
AI4I repeated experimental framework	Complete
Five deployment-dimension evidence structure	Complete
Candidate-level and repetition-level evidence construction	Complete
Tie-aware criterion-ranking functionality	Complete
LOO empirical reliability estimation	Complete
Reliability-weight construction	Complete
Primary RDRA consensus implementation	Complete
Equal Borda and Existing Composite baselines	Complete
Dominance-consistency sanity testing	Complete
VQ1 criterion-uncertainty validation	Complete

Reliability-weight sensitivity analysis	Complete
VQ2 data-quality responsiveness analysis	Complete
VQ3 preference-uncertainty analysis	Complete
Exploratory reliability-influence refinement	Complete and locked for subsequent evaluation
Complementary C-MAPSS evaluation	Complete within the defined scope
Final Research Implementation consolidation	Ongoing

The status demonstrates the broad completion of the planned research work expected by the task. The remaining work is primarily concerned with report consolidation and the deeper interpretation of the completed findings in the later Research Analysis stage rather than development of the core RDRA functionality.

4.4 Scope Limitations and Trade-Offs

The research scope was deliberately kept focused on the core RDRA contribution. AI4I provides the repeated five-dimension evidence required for full reliability-aware development and validation [9], while C-MAPSS provides complementary evidence from a different predictive-maintenance task [22], [23]. This keeps the implementation manageable while still providing experimental breadth.

The five AI4I deployment dimensions represent the criteria operationalised in this study and are not intended to cover every possible industrial deployment consideration. Similarly, the resulting reliability weights should be interpreted as empirical ranking-reproducibility estimates under the implemented experimental protocol, rather than as universal criterion weights or stakeholder priorities.

The use of rank-based aggregation also represents a deliberate trade-off. It provides a common basis for combining heterogeneous deployment evidence but focuses on relative candidate ordering rather than preserving every absolute difference between raw metric values. This is appropriate to the research because the proposed contribution specifically concerns criterion-ranking reproducibility and consensus formation.

Finally, the exploratory reliability-influence refinement remains separate from primary RDRA, and the C-MAPSS branch remains complementary rather than a second complete replication. These boundaries preserve a clear distinction between the primary contribution, exploratory development and supporting cross-task evidence while avoiding claims beyond what the implemented research currently supports.

5. Solution Development and Implementation

The conceptual RDRA solution defined in Section 3 was operationalised as a modular Python/Google Colab computational research artefact. The implementation transforms experimentally generated predictive-maintenance evidence into deployment-oriented criterion scores, preserves repetition-level information for empirical reliability estimation, produces criterion-specific rankings, estimates ranking reproducibility, constructs reliability weights and generates the final reliability-aware deployment consensus.

The implementation was deliberately modular rather than being implemented as a single function that accepts experimental data and returns only a final ranking. Intermediate outputs are retained at each major stage, including deployment-dimension scores, criterion-specific ranks, repetition-level scores, Leave-One-Repetition-Out (LOO) agreement values, empirical stability estimates, reliability weights, Borda utilities and final consensus scores. This design supports traceability because an unexpected final ranking can be examined backwards through the evidence and transformations that produced it.

The primary technical implementation uses AI4I 2020 [6], which provides the repeated five-dimension evidence required for complete RDRA reliability estimation. C-MAPSS FD001 [22], [23] is implemented separately as complementary cross-task evidence and is evaluated later within its defined scope.

5.1 Development Plan

Development was organised as a sequence of dependent activities. Experimental evidence was prepared before the reliability-aware mechanism was introduced, and scientific validation was implemented after the primary RDRA artefact had been defined. This ordering reduced the risk of adapting the proposed method retrospectively around favourable evaluation results.

Table 5.1 Detailed RDRA solution-development plan

Development task	Key activities	Resources required	Tangible output
1. Prepare experimental evidence	Load candidate-level and repetition-level predictive, robustness, calibration, efficiency and operational evidence	AI4I outputs, Python, pandas	Structured experimental evidence
2. Construct deployment dimensions	Align benefit/cost directions and combine related metrics	pandas, NumPy	Five deployment-dimension scores

3. Generate criterion rankings	Rank candidates independently and preserve tied positions	pandas	Criterion-specific rank table
4. Preserve repetition-level evidence	Construct the same five dimensions separately for each experimental seed	AI4I repeated outputs, pandas	Seed × candidate × dimension evidence
5. Estimate empirical reliability	Perform LOO reference construction and Kendall rank comparison	SciPy, pandas, NumPy	Criterion stability estimates
6. Construct reliability weights	Clip negative agreement, average positive stability and normalise criterion influence	NumPy	RDRA reliability-weight vector
7. Implement Borda aggregation	Convert ranks to normalised Borda utilities and calculate Equal Borda and RDRA consensus scores	pandas, NumPy	Baseline and RDRA rankings
8. Implement exploratory refinement	Apply the predefined γ transformation using development-only VQ1 evidence	NumPy, itertools	Locked exploratory refined RDRA
9. Preserve implementation artefacts	Export stability, weighting, ranking and development outputs	pathlib, pandas, Matplotlib	CSV files, tables and figures
10. Scientifically evaluate the solution	Perform VQ1, VQ2 and VQ3 testing	SciPy, NumPy, Matplotlib	Scientific evidence presented in Section 6

5.1.1 Key Development Tasks and Activities

The development activities follow a clear dependency structure:

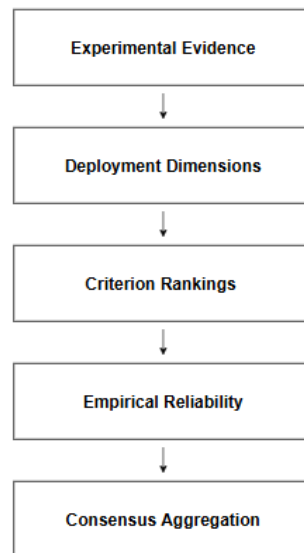


Figure 5.1 Sequential process from experimental evidence to consensus aggregation

Deployment dimensions must be constructed before ranking because the underlying measurements use different numerical scales and optimisation directions. Criterion rankings must then exist before their reproducibility can be estimated. Reliability weights are subsequently required before the proposed RDRA consensus can be generated.

The scientific-evaluation modules were implemented after the primary artefact had been defined. This sequencing is methodologically important because VQ1-VQ3 should evaluate an established proposed method rather than be repeatedly used to redefine the core algorithm until favourable outcomes are obtained.

The exploratory γ -based refinement was similarly isolated from later validation. Its development used only designated development evidence, after which the selected configuration was locked before VQ2 and VQ3 interpretation.

5.1.2 Required Resources and Research Artefacts

The implementation operates on experimental artefacts generated during the earlier predictive-maintenance experiments rather than on hypothetical model scores. For AI4I, these include candidate-level summary evidence, clean repetition-level measurements and repetition-level corruption-robustness evidence.

The different artefacts serve distinct functions. Candidate-level summaries provide the evidence required to construct the final deployment rankings, whereas repetition-level outputs preserve the variation required by the LOO reliability estimator.

The main computational resources required were Pyhon, Google Colab, pandas, NumPy, SciPy, Matplotlib, scikit-learn, XGBoost, TensorFlow/Keras, pathlib, itertools and structured CSV outputs from the experimental stages.

Reproducibility controls were incorporated into the computational workflow. A fixed project random seed is used for procedures involving random sampling:

```
RNG_SEED = 20260811
rng = np.random.default_rng(RNG_SEED)
```

This ensures that procedures such as exploratory development splitting and preference-scenario generation can be reproduced rather than depending on an uncontrolled random state.

Input artefacts are also checked explicitly before processing:

```
def require(path):
    path = Path(path)
    if not path.exists():
        raise FileNotFoundError(
            f"Required output not found: {path}\n"
            "Run the relevant earlier notebook first."
        )
    return path
```

This prevents the aggregation pipeline from silently continuing when an upstream experimental artefact is absent. Thus, the implementation treats data availability and reproducibility as explicit parts of the computational workflow.

5.2 Technology Stack and Tool Selection

The implementation uses a lightweight scientific-computing stack capable of supporting evidence processing, machine-learning outputs, statistical analysis, ranking and visualisation within one environment.

Table 5.2 Technology stack used for solution development

Technology / tool	Role in the solution	Justification
Python	Primary implementation language	Integrates machine learning, numerical computation, statistics and data processing
pandas	Candidate/repetition evidence manipulation	Appropriate for structured tabular data and ranking operations
NumPy	Numerical transformation and aggregation	Supports efficient vector and matrix operations

SciPy	Kendall and Spearman rank agreement	Provides established statistical implementations
Matplotlib	Research visualisation	Generates figures directly from computational outputs
scikit-learn	AI4I model development and evaluation	Supports conventional ML and metric calculation
XGBoost	Gradient-boosting candidate models	Provides an additional strong tree-based candidate family
TensorFlow/Keras	C-MAPSS sequential models	Supports LSTM and GRU implementation
pathlib	Input/output management	Provides explicit and portable artefact paths
itertools	Criterion-combination generation	Supports VQ1 and exploratory development procedures

Python was selected because the complete research pipeline can be maintained within one programming language. This limits unnecessary data transfer between software platforms and makes the transformation from experimental output to ranking evidence easier to reproduce.

Google Colab was selected as the notebook-based development environment because executable Python code can be combined with mathematical definitions, tabular outputs and generated figures. This is especially appropriate for a computational research prototype where both the implementation and the evidence produced by that implementation must remain inspectable.

The machine-learning libraries primarily support the upstream experimental stages. The RDRA aggregation mechanism itself does not depend on the internal structure of any specific family model, reinforcing its role as a deployment-selection layer rather than a predictive algorithm.

5.3 Implementation of Core RDRA Functionalities

The core implementation translates the mathematical formulation in Section 3.2 into an executable computational pathway. Each major methodological component is implemented explicitly so that intermediate outputs can be inspected independently.

The primary functions are:

1. deployment-dimension construction
2. criterion-specific ranking
3. normalised Borda transformation
4. LOO empirical reliability estimation
5. reliability-weight construction
6. Equal Borda and RDRA aggregation
7. exploratory reliability-influence refinement

5.3.1 Deployment-Dimension Construction

The first implementation stage converts heterogeneous experimental measurements into consistently oriented deployment scores.

Some metrics are naturally maximised. For example, larger F1-score and PR-AUC values represent stronger predictive behaviour. Other metrics are naturally minimised, including Brier score, Expected Calibration Error, inference latency, model size and operational cost.

Therefore, two transformation functions were implemented:

```
def minmax_benefit(series):
    s = pd.Series(series, dtype=float)
    lo, hi = s.min(), s.max()
    if np.isclose(lo, hi):
        return pd.Series(np.ones(len(s)), index=s.index, dtype=float)
    return (s - lo) / (hi - lo)

def minmax_cost(series):
    return 1.0 - minmax_benefit(series)
```

minmax_benefit() maps observed values onto a common interval where larger values indicate more desirable evidence. minmax_cost() reverses this orientation for metrics where smaller values are preferable.

The equal-value safeguard is important. If every candidate has effectively the same value for a metric, conventional min-max normalisation would divide by zero. Returning equal scores instead correctly represents the fact that the metric provides no discrimination between the candidates. After transformation, all dimensions follow the same interpretation: a higher score indicates more desirable deployment behaviour.

The five deployment dimensions are represented in code as:

```
AI_DIMS = [
    "Predictive",
    "Robustness",
    "Reliability",
    "Efficiency",
    "Operational",
]
```

The shorter implementation label Reliability corresponds to the **Probability Reliability** dimension used in the report.

The candidate-level dimensions are constructed using

```

ai_scores["Predictive"] = (
    minmax_benefit(ai_candidate["F1"])
    + minmax_benefit(ai_candidate["PR_AUC"])
) / 2.0

ai_scores["Robustness"] = minmax_benefit(
    ai_candidate["Composite_Corruption_Robustness_Mean"]
)

ai_scores["Reliability"] = (
    minmax_cost(ai_candidate["Brier"])
    + minmax_cost(ai_candidate["ECE"])
) / 2.0

ai_scores["Efficiency"] = (
    minmax_cost(ai_candidate["Single_Sample_Latency_Median_ms"])
    + minmax_cost(ai_candidate["Model_Size_MB"])
) / 2.0

ai_scores["Operational"] = minmax_cost(
    ai_candidate["Operational_Cost"]
)

```

The code operationalises the five deployment dimensions defined in Section 2.1 while keeping them separate until aggregation. This separation is necessary because the research investigates disagreement among deployment perspectives, prematurely collapsing the dimensions into one score would remove the criterion-specific rankings required by RDRA.

5.3.2 Criterion Ranking and Normalised Borda Transformation

Once all five deployment dimensions use the same benefit-oriented direction, each criterion independently generates a candidate ranking. The ranking function is:

```

def rank_from_score(series):
    return pd.Series(series).rank(
        ascending=False,
        method="average")

```

`ascending=False` ensures that the largest deployment score receives rank 1. The use of `method="average"` ensures that tied candidates receive the same average position. This prevents the software from introducing arbitrary ordering that is unsupported by experimental evidence.

The ranks are subsequently transformed using the normalised Borda expression defined in Section 3.2.5:

$$B_{ij} = \frac{n - r_{ij}}{n - 1}.$$

The implementation is:

```

def normalized_borda_from_rank(rank_series):
    r = pd.Series(rank_series, dtype=float)
    n = len(r)
    if n <= 1:
        return pd.Series(
            np.ones(n),
            index=r.index,
            dtype=float)
    return (n - r) / (n - 1)

```

Floating-point rank values are deliberately preserved so that fractional average ranks remain valid. The $n \leq 1$ safeguard also prevents division by zero if the function is ever applied to a degenerate one-candidate input.

The Borda transformation then places all criterion rankings onto the same ordinal utility scale while preserving their relative ordering. The contribution of RDRA is not this transformation itself, Borda-style aggregation is established [16], but the empirical reliability information subsequently applied to these criterion utilities.

5.3.3 LOO Empirical Reliability Estimation

The most important distinctive functionality of RDRA is the estimation of criterion-level empirical ranking reproducibility. The purpose of this stage is to determine whether each deployment criterion produces a similar candidate ordering across the repeated AI4I experiments.

This stage requires the five deployment dimensions to be reconstructed separately for each experimental repetition rather than only at the candidate-average level. The clean repetition-level evidence is merged with repetition-specific robustness evidence:

```
seed_base = ai_clean.merge(
    ai_composite_seed[
        ["Seed", "Variant", "Composite_Corruption_Robustness"]
    ],
    on=["Seed", "Variant"],
    how="left",
)
```

The deployment dimensions are then reconstructed for each seed:

```
seed_score_rows = []
for seed, g in seed_base.groupby("Seed"):
    g = g.copy()

    seed_score_rows.append(pd.DataFrame({
        "Seed": seed,
        "Variant": g["Variant"].to_numpy(),
        "Predictive": (
            minmax_benefit(g["F1"]).to_numpy()
            + minmax_benefit(g["PR_AUC"]).to_numpy()
        ) / 2.0,
        "Robustness": minmax_benefit(
            g["Composite_corruption_Robustness"]
        ).to_numpy(),
        "Reliability": (
            minmax_cost(g["Brier"]).to_numpy()
            + minmax_cost(g["ECE"]).to_numpy()
        ) / 2.0,
        "Efficiency": (
            minmax_cost(
                g["Single_Sample_Latency_Median_ms"]
            ).to_numpy()
            + minmax_cost(g["Model_Size_MB"]).to_numpy()
        ) / 2.0,
        "Operational": minmax_cost(
            g["Operational_Cost"]
        ).to_numpy(),
    }))

ai_seed_dimension_scores = pd.concat(
    seed_score_rows,
    ignore_index=True,
)
```

This produces a repeated evidence structure of the form:

Repetition × *Candidate* × *Deployment Dimension*

This structure is essential because averaging the repetitions before the reliability stage would remove the information required to determine whether the candidate ordering is reproducibly supported.

The LOO estimator then evaluates each criterion separately. For every repetition, its candidate ranking is compared with a reference ranking generated from the remaining repetitions using Kendall agreement.

Conceptually:

$$\tau_{jr} = \tau(r_j^{(r)}, r_j^{(-r)}).$$

The stability procedure is applied using:

```
ai_stability = estimate_dimension_stability_loo(  
    ai_seed_dimension_scores,  
    "Variant",  
    "Seed",  
    AI_DIMS,  
)
```

The implementation retains both the signed mean LOO agreement and the positive reproducibility evidence used by RDRA. The latter is calculated as:

```
positive_stability = (  
    float(np.mean(np.clip(tau_values, 0.0, 1.0)))  
    if tau_values else 1.0  
)
```

This implements:

$$s_j = \frac{1}{R} \sum_{r=1}^R \max(0, \tau_{jr}).$$

Hence, negative agreement contributes no positive reliability evidence rather than creating a negative aggregation weight.

This distinction is deliberate. RDRA is intended to reduce the influence of poorly reproducible criterion evidence. It is not intended to reverse the meaning of a deployment criterion. Allowing a negative stability value to become a negative weight could cause strong performance under a criterion to reduce a candidate's final score, which would represent criterion inversion rather than low confidence.

The implementation also handles the edge case in which no valid LOO comparisons are available:

```

mean_tau = (
    float(np.mean(tau_values))
    if tau_values else 1.0
)

# Reliability cannot be negative.
# Negative rank agreement is treated as zero evidence reliability.
positive_stability = (
    float(np.mean(np.clip(tau_values, 0.0, 1.0)))
    if tau_values else 1.0
)

rows.append({
    "Dimension": dim,
    "Mean_Kendall_Tau_LOO": mean_tau,
    "Stability_Weight_Raw": positive_stability,
    "N_Repetitions": len(tau_values),
})

```

Assigning zero stability in this situation follows the conservative principle that absence of reliability evidence does not equal evidence of perfect reliability. This safeguard does not affect the reported AI4I results because valid LOO comparisons are available across the 20 repetitions, but it provides a logically consistent behaviour for future incomplete evidence.

5.3.4 Reliability-Weight Construction

The raw positive stability values produced by the LOO procedure must be converted into a valid criterion-weight vector. The implementation calculates:

```

total = stability_df["Stability_Weight_Raw"].sum()

if np.isclose(total, 0):
    stability_df["Reliability_Weight"] = (
        1.0 / len(stability_df)
    )
else:
    stability_df["Reliability_Weight"] = (
        stability_df["Stability_Weight_Raw"] / total
    )

```

This operationalises:

$$w_j = \frac{s_j}{\sum_k s_k}.$$

The zero-total fallback is important because if none of the criteria contains positive reproducibility evidence, there is no empirical basis for assigning unequal influence. Under that condition the method reduces naturally to Equal Borda:

$$w_j = \frac{1}{m}.$$

The resulting AI4I reliability-weight dictionary is created using:

```

ai_reliability_weights = dict(zip(
    ai_stability["Dimension"],
    ai_stability["Reliability_Weight"],
))

```

The actual output from the 20 AI4I repetitions is shown in Table 5.3.

Table 5.3 LOO empirical reliability estimates from the primary AI4I implementation

Deployment dimension	Mean Kendall Tau LOO	Positive stability S_j	Repetitions	Reliability weight w_j
Predictive Quality	0.651369	0.651369	20	0.222560
Robustness	0.376062	0.380964	20	0.130168
Probability Reliability	0.788889	0.788889	20	0.269549
Computational Efficiency	0.818301	0.818301	20	0.279598
Operational Impact	0.258432	0.287183	20	0.098125

The output confirms that the criterion rankings are not equally reproducible under the implemented experimental protocol. Computational Efficiency and Probability Reliability receive larger empirical reliability weights, whereas Operational Impact and Robustness receive smaller values.

These values must not be interpreted as stakeholder importance. For instance, $w_{\text{Efficiency}} > w_{\text{Operational}}$ does not mean that Computational Efficiency is inherently more important than Operational Impact. It means only that the Efficiency candidate ordering was reproduced more consistently across the 20 AI4I repetitions.

The difference between Mean_Kendall_Tau_LOO and Stability_Weight_Raw for Robustness and Operational Impact is also expected. The former averages signed Kendall values, whereas the latter clips negative repetition-level agreements to zero before averaging $\text{mean}(\tau) \neq \text{mean}(\max(0, \tau))$ when negative values are present.

Both quantities are retained so that the reliability transformation remains inspectable.

The output is exported as a persistent computational artefact:

```
ai_stability.to_csv(  
    OUT / "01_ai4i_dimension_reliability_weights_LOO.csv",  
    index=False,  
)
```

5.3.5 Equal Borda and RDRA Consensus Aggregation

The final primary functionality combines the criterion-specific Borda utilities into a single deployment-oriented consensus ranking. A deliberate implementation choice is that Equal Borda and RDRA use the same aggregation function:

```
def aggregate_borda(
    rank_df,
    id_col,
    dimensions,
    reliability_weights=None,
    aggregation_name="RDRA",
):
    if reliability_weights is None:
        reliability_weights = {d: 1.0 for d in dimensions}

    weights = np.array(
        [max(float(reliability_weights.get(d, 0.0)), 0.0) for d in dimensions],
        dtype=float,
    )

    if np.isclose(weights.sum(), 0):
        weights = np.ones(len(dimensions), dtype=float)

    weights = weights / weights.sum()

    utility_matrix = np.column_stack([
        normalized_borda_from_rank(rank_df[f"{d}_Rank"].to_numpy())
        for d in dimensions
    ])

    aggregate = utility_matrix @ weights

    result = rank_df.copy()
    result[f"{aggregation_name}_Score"] = aggregate
    result[f"{aggregation_name}_Rank"] = pd.Series(
        aggregate, index=result.index
    ).rank(
        ascending=False,
        method="average"
    )

    return result, dict(zip(dimensions, weights))
```

The function first determines the criterion-weight vector. If `reliability_weights=None`, every criterion initially receives the same value. Normalising these values produces:

$$w_j = \frac{1}{m}$$

With five dimensions, each criterion therefore receives a weight of 0.20: $w_j = 0.20$.

When the empirical reliability dictionary from Section 5.3.4 is supplied, the same function performs primary RDRA aggregation.

The final candidate score implements:

$$S_i = \sum_{j=1}^m w_j B_{ij}.$$

The resulting aggregate scores are converted to final ranks using the same tie-aware logic:

```

result = rank_df.copy()
result[f"{aggregation_name}_Score"] = aggregate
result[f"{aggregation_name}_Rank"] = pd.Series(
    aggregate, index=result.index
).rank(
    ascending=False,
    method="average"
)

```

The methods are generated through:

```

equal_borda, equal_borda_weights = aggregate_borda(
    ai_dimension_ranks,
    "Variant",
    AI_DIMS,
    None,
    "Equal_Borda",
)

rdra, rdra_weights = aggregate_borda(
    ai_dimension_ranks,
    "Variant",
    AI_DIMS,
    ai_reliability_weights,
    "RDRA",
)

```

This shared implementation provides an important experimental control. Equal Borda and RDRA use the same:

- candidate configuration
- deployment dimensions
- criterion rankings
- tie-handling procedure
- Borda transformation
- utility matrix
- final score-to-rank procedure.

Therefore, the substantive difference is the criterion-weight vector:

$$w^B = [0.20, , 0.20, , 0.20 \ 0.20 \ 0.20]$$

Versus the empirical primary RDRA vector:

$$w^{RDRA} \approx [0.223, , 0.130, , 0.270 \ 0.280 \ 0.098]$$

This isolates empirical reliability weighting as the substantive difference between the methods. If separate aggregation procedures had been implemented, differences in the resulting rankings could potentially arise from unrelated coding choices.

An Existing Composite Method is also retained as an additional reference:

```

existing = ai_scores.copy()
existing["Existing_Composite_Score"] = existing[
    AI_DIMS
].mean(axis=1)
existing["Existing_Composite_Rank"] = existing[
    "Existing_Composite_Score"
].rank(
    ascending=False,
    method="average",
)

```

The Existing Composite is not part of the RDRA mechanism. Its role is to provide a broader simple-aggregation reference alongside the controlled Equal Borda baseline.

5.3.6 Exploratory Reliability-Influence Refinement

The primary RDRA method uses reliability weights directly proportional to the positive LOO stability estimates:

$$w_j = \frac{s_j}{\sum_k s_k},$$

which corresponds to a reliability-influence exponent of:

$$\gamma = 1.$$

After the primary method had been implemented, a controlled exploratory extension was added to investigate whether the strength of reliability influence affects ranking behaviour. The function is:

```

def make_gamma_reliability_weights(
    stability_df,
    dimensions,
    gamma=1.0,
    stability_col="Stability_Weight_Raw",
    eps=1e-12,
):
    stability_lookup = dict(zip(
        stability_df["Dimension"],
        stability_df[stability_col],
    ))

    raw = np.array(
        [
            max(
                float(stability_lookup[d]),
                eps,
            ) ** float(gamma)
            for d in dimensions
        ],
        dtype=float,
    )

    if np.isclose(raw.sum(), 0.0):
        raw = np.ones(len(dimensions), dtype=float)

    raw = raw / raw.sum()

    return dict(zip(dimensions, raw))

```

This implements:

$$w_j(\gamma) = \frac{s_j^\gamma}{\sum_k s_k^\gamma}.$$

The conceptual meaning of γ was defined in Section 3.2.8. Within the implementation, the key issue is how the exploratory value was selected without allowing later validation results to influence that selection.

All genuine VQ1 criterion subsets of sizes two, three and four were first generated:

```
all_vq1_subsets = []

for k in range(2, len(AI_DIMS)):
    for dims_tuple in itertools.combinations(AI_DIMS, k):
        all_vq1_subsets.append(tuple(dims_tuple))

assert len(all_vq1_subsets) == 25
```

This corresponds to:

$$\binom{5}{2} + \binom{5}{3} + \binom{5}{4} = 10 + 10 + 5 = 25$$

The complete five-criterion ranking is retained as the reference and is not counted as a 26th perturbation, because including the reference as its own perturbation would introduce an artificial perfect-agreement observation.

The subset scenarios were shuffled using the fixed project random state and divided approximately 60/40 into exploratory development and held-out groups.

The predefined search grid was:

```
GAMMA_GRID = [
    0.0,
    0.25,
    0.50,
    0.75,
    1.0,
    1.25,
    1.50,
    2.0,
    3.0,
]
```

The development procedure selected:

$$\gamma = 3.0.$$

Once selected, this value was locked before VQ2 and VQ3 evaluation. This is an important methodological safeguard because changing γ after inspecting later validation results would use the same evidence both to refine and evaluate the method, increasing the risk of validation leakage.

The selected value should nevertheless be interpreted cautiously. Because:

$$\gamma = 3.0$$

is the upper boundary of the predefined grid, the experiment does not establish that $\gamma = 3$ is a globally optimal reliability-influence parameter. A wider search range could potentially produce a different development outcome. Consequently, the research maintains a strict distinction between **Primary RDRA**: $\gamma = 1$, representing the original reliability-proportional formulation and **Locked exploratory refined RDRA**: $\gamma = 3$, representing a development-selected extension used to investigate stronger reliability influence.

5.4 Implementation Outputs and Functional Artefacts

The implementation produces persistent intermediate and final outputs so that the complete computational path from experimental evidence to deployment consensus remains inspectable.

These artefacts provide evidence that the conceptual method has been implemented as a functioning prototype rather than remaining only a mathematical proposal.

5.4.1 Implemented Dataflow Artefact

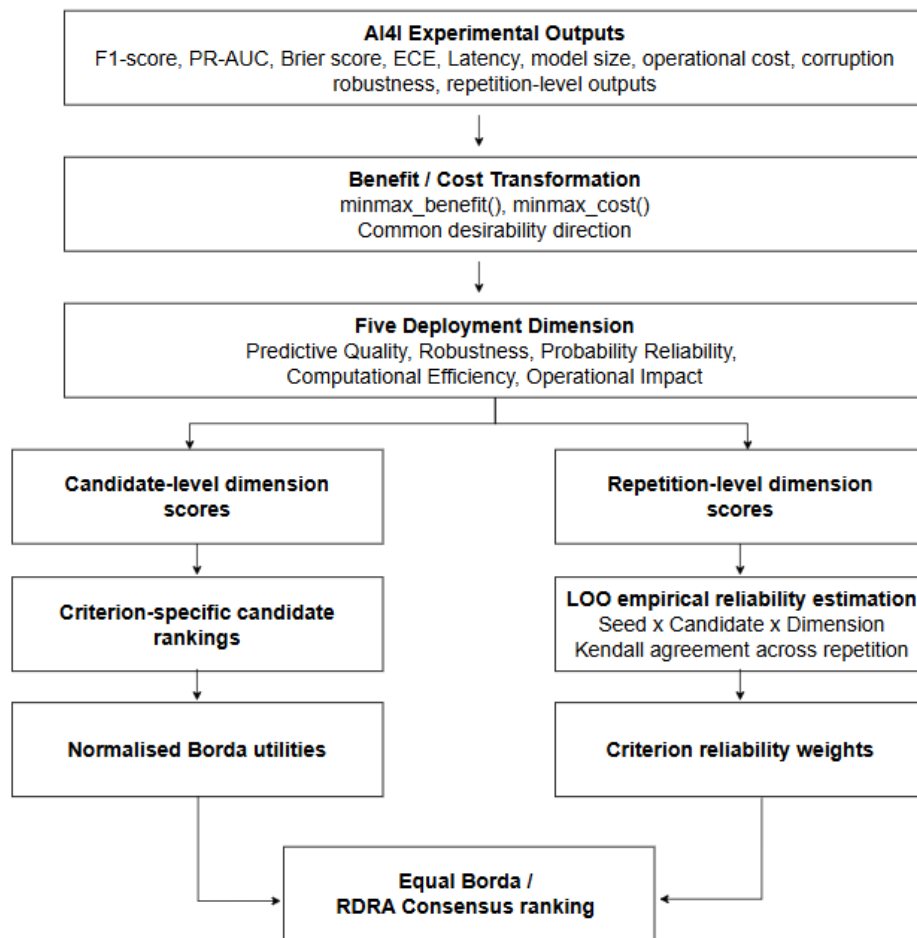


Figure 5.2 Implemented dataflow from experimental evidence to deployment-dimension evidence

Figure 3 represents the operational pathway through which heterogeneous AI4I outputs are transformed into consistently oriented deployment dimensions and separated into candidate-level and repetition-level evidence.

The candidate-level branch supports final criterion ranking and consensus aggregation, while the repetition-level branch supports empirical reliability estimation. The artefact then demonstrates how the implementation preserves the evidence required by both the final ranking and the LOO reliability mechanism.

5.4.2 Empirical Reliability Artefact

	Dimension	Mean_Kendall_Tau_LOO	Stability_Weight_Raw	N_Repetitions	Reliability_Weight
0	Predictive	0.651369	0.651369	20	0.222560
1	Robustness	0.376062	0.380964	20	0.130168
2	Reliability	0.788889	0.788889	20	0.269549
3	Efficiency	0.818301	0.818301	20	0.279598
4	Operational	0.258432	0.287183	20	0.098125

RDRA reliability weights:
{'Predictive': 0.22256045180661224, 'Robustness': 0.1301681563778519, 'Reliability': 0.26954850291200555, 'Efficiency': 0.2795979499965459, 'Operational': 0.09812493890698448}

Figure 5.3 LOO criterion-ranking stability and empirical RDRA reliability weights across 20 AI4I repetitions

The output shows the main functional output of the reliability-estimation component. The implementation successfully generates distinct criterion-level stability and weight values rather than assuming $w_j = 0.20$ for every criterion.

The resulting primary RDRA weight vector is approximately:

$$w^{RDRA} \approx [0.223, 0.130, 0.270, 0.280, 0.098].$$

This output provides tangible evidence that the central proposed functionality, conversion of repeated ranking reproducibility into aggregation influence, has been implemented.

5.4.3 Traceability and Reproducibility

Intermediate and final outputs are preserved as structured computational artefacts rather than existing only in notebook memory.

The exported outputs include:

- deployment-dimension evidence
- criterion-specific rankings
- LOO stability estimates
- empirical reliability weights
- Equal Borda outputs
- primary RDRA outputs

- exploratory-development output
- later scientific-evaluation tables and figures.

This makes the research process traceable from experimental measurements through to the final deployment consensus and supports subsequent analysis without requiring the complete predictive-model training pipeline to be rerun for every aggregation analysis.

5.5 Section Summary

Google Colab research artefact operationalises the complete RDRA pathway from heterogeneous predictive-maintenance evidence to reliability-aware deployment consensus. The implementation constructs five consistently oriented deployment dimensions, generates tie-aware criterion rankings, transforms rankings into normalised Borda utilities, estimates empirical criterion-ranking reproducibility through LOO Kendall agreement, converts positive stability into reliability weights and generates both Equal Borda and primary RDRA consensus rankings through a shared aggregation procedure.

The implementation also includes conservative safeguards for equal-valued metrics, missing reliability evidence and zero total stability, together with persistent outputs and fixed random-state controls that improve reproducibility and traceability. The exploratory γ -based extension was implemented separately from primary RDRA and selected using development-only evidence before being locked for subsequent evaluation.

Therefore, these components demonstrate that the proposed conceptual solution has progressed into a functioning computational research artefact satisfying the implementation requirements established in Section 4. Section 6 subsequently evaluates whether this implemented reliability-aware mechanism provides scientifically measurable value under criterion uncertainty, reliability-weight uncertainty, data-quality uncertainty and stakeholder-preference uncertainty.

6. Solution Evaluation and Scientific Testing

The implemented RDRA solution was evaluated at two complementary levels: **functional verification** and **scientific evaluation**. Functional verification determines whether the computational artefact performs the capabilities specified in Section 4 and implemented in Section 5. Scientific evaluation then determines whether the reliability-aware ranking mechanism exhibits the behavioural properties for which it was designed.

This distinction is essential: the mere fact that an artefact functions correctly does not establish that it offers any scientifically validated advantage. Functional success reflects operational capability, whereas scientific value requires evidence-based improvement.

Therefore, the evaluation does not assume that RDRA must outperform Equal Borda on every measure. Instead, it examines whether empirical reliability-aware aggregation provides measurable value under specific forms of deployment uncertainty.

The scientific testing addresses three complementary validation questions:

1. **VQ1 - Criterion Uncertainty:** Does primary RDRA maintain a more consistent consensus when the available deployment criteria change?
2. **VQ2 - Data-Quality Uncertainty:** Does the ranking respond meaningfully when the supporting deployment evidence changes?
3. **VQ3 - Preference Uncertainty:** Does the resulting shortlist remain useful when stakeholder deployment priorities vary?

These questions correspond to the consistency, responsiveness and downstream-usefulness properties motivated in Section 2.3.

6.1 Evaluation Strategy and Requirement Achievement

6.1.1 Evaluation Objectives

The evaluation framework was designed to examine the implemented solution from several complementary perspectives.

First, functional verification determines whether the required evidence-processing, ranking, reliability-estimation and aggregation components operate successfully. Second, evidence-alignment testing provides a basic sanity check that the resulting consensus does not contradict straightforward dominance relationships in the deployment evidence.

The scientific evaluation then proceeds through VQ1 Criterion Uncertainty to assess ranking consistency. Reliability-Weight Sensitivity to assess internal robustness to uncertainty in the estimated weights, VQ2 Data-Quality Uncertainty to assess responsiveness to meaningful evidence deterioration, and VQ3 Preference Uncertainty to examine downstream shortlist usefulness. Finally, C-MAPSS provides complementary cross-task evidence in a substantially different predictive-maintenance setting.

The evaluation begins by confirming whether the artefact functions correctly. It then checks whether the outputs are aligned with the underlying deployment evidence. Next, it assesses whether the ranking remains consistent under changes in available criteria. The process continues by examining whether the method is robust to uncertainty in its own reliability weights. It then evaluates whether the ranking responds appropriately when the deployment evidence meaningfully changes. Finally, it determines whether the resulting Top-3 shortlist is useful for downstream decision-making.

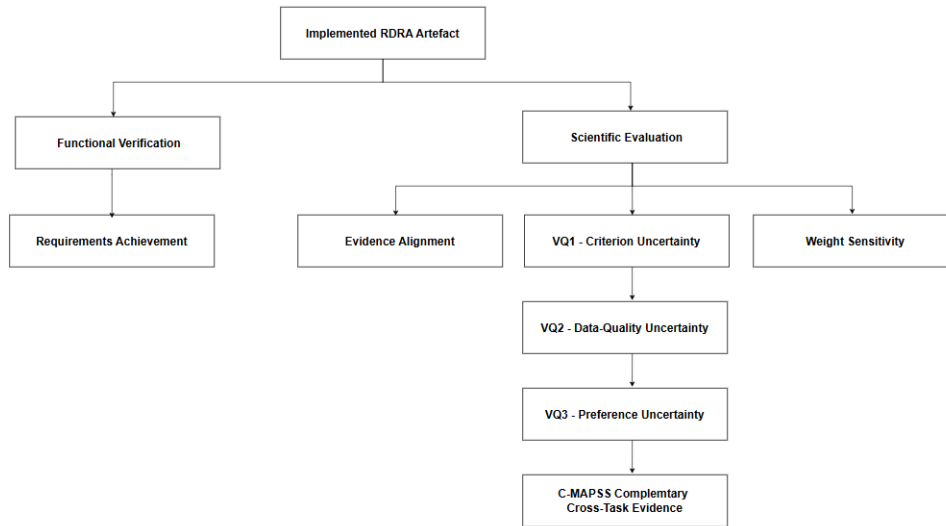


Figure 6.1 Scientific evaluation framework for the implemented RDRA solution

Figure 6.1 represents the organisation of the evaluation rather than a sequence in which every later test is expected to demonstrate superiority.

6.1.2 Functional and Requirement Verification

The first evaluation stage verifies the requirements defined in Section 4.

Table 6.1 Verification of major RDRA requirements

Requirement group	Evaluation evidence	Assessment
FR1-FR4: evidence preparation and dimension construction	Candidate- and repetition-level evidence successfully transformed into five deployment dimensions	Functionally verified
FR5-FR6: criterion ranking and tie handling	Separate tie-aware rankings generated for all five dimensions	Functionally verified
FR7-FR8: LOO reliability estimation and weight construction	LOO agreement calculated across 20 repetitions and converted to empirical reliability weights	Functionally verified
FR9-FR12: consensus and comparison methods	Normalised Borda, Equal Borda, primary RDRA and Existing Composite outputs generated	Functionally verified
FR13: evidence alignment	Dominance-consistency sanity test completed	Verified through sanity testing

FR14: criterion uncertainty	25 genuine criterion-subset perturbations evaluated	Scientifically evaluated through VQ1
FR15: reliability-weight sensitivity	1,000 controlled perturbations evaluated	Scientifically evaluated
FR16: data-quality uncertainty	16 controlled AI4I corruption scenarios evaluated	Scientifically evaluated through VQ2
FR17: stakeholder-preference uncertainty	5,000 matched preference scenarios evaluated	Scientifically evaluated through VQ3
FR18: traceability	Intermediate and final tables and figures exported	Functionally verified
FR19: primary/exploratory separation	Primary $\gamma = 1$ retained separately from locked exploratory $\gamma = 3$	Methodologically verified
FR20: complementary cross-task evidence	C-MAPSS FD001 evaluation implemented	Verified within defined complementary scope

The verification confirms that the proposed solution has progressed from conceptual design to an executable research artefact and that the functionality required for subsequent scientific evaluation is present. The remaining subsections evaluate whether this functioning artefact exhibits the properties targeted by the research.

6.2 Evidence-Alignment and Dominance Sanity Testing

Before evaluating uncertainty behaviour, a dominance-consistency sanity test was performed to determine whether the aggregated ranking contradicted straightforward relationships in the deployment evidence.

A candidate was treated as dominating another when it was at least as favourable across the comparable deployment dimensions and strictly more favourable on at least one dimension. A dominance violation would occur if the consensus ranked the dominated candidate above its dominating alternative.

The analysis identified 9 comparable dominance pairs and 0 violations. Therefore,

$$\text{Dominance Consistency} = \frac{9 - 0}{9} = 1.0.$$

The resulting consensus was consistent with all directly comparable dominance relationships identified in the current evidence.

This result is interpreted as a sanity check. Dominance consistency establishes that the implemented aggregation does not contradict obvious deployment-evidence relationships. The more demanding tests of ranking behaviour are provided by VQ1-VQ3.

6.3 VQ1: Evaluation under Criterion Uncertainty

VQ1 evaluates whether primary RDRA maintains a reasonably consistent deployment consensus when some criteria are unavailable.

Criterion availability can vary in practice because complete deployment evidence may not always be available at a particular model-selection stage. Therefore, the purpose of VQ1 is not to require an identical ranking after criterion removal, but to determine whether ranking changes remain proportionate rather than unnecessarily disruptive.

6.3.1 Experimental Design

With five deployment dimensions, all criterion subsets containing two, three or four dimensions were generated:

$$\binom{5}{2} + \binom{5}{3} + \binom{5}{4} = 10 + 10 + 5 = 25.$$

Thus, 25 genuine criterion-availability perturbations were evaluated.

The complete five-criterion ranking was retained as the reference. It was **not** included as a 26th perturbation because comparison of the reference ranking with itself would introduce an artificial perfect-agreement observation and inflate the consistency result.

Three aggregation methods were examined:

- Existing Composite
- Equal Borda
- Primary RDRA

The principal consistency measures were:

- Kendall rank agreement
- Spearman rank correlation
- Top-3 shortlist overlap
- mean absolute rank displacement
- tie-aware winner survival
- Top-1 set overlap

Hence, these measures distinguish broader ranking consistency from exact-winner stability.

6.3.2 Descriptive VQ1 Results

	Method	Mean_Kendall	Mean_Spearman	Mean_Top3_Overlap	Mean_Absolute_Rank_Displacement	Tie_Aware_Winner_Survival	Mean_Top1_Set_Overlap	N_Perturbations
0	Equal_Borda	0.677306	0.799260	0.720000	2.257778	0.32	0.20	25
1	Existing_Composite	0.624052	0.741754	0.746667	2.448889	0.36	0.36	25
2	RDRA	0.746405	0.862456	0.813333	1.773333	0.36	0.36	25

Figure 6.2 VQ1 descriptive consistency over 25 criterion perturbations

Based on the result, primary RDRA produced the highest mean Kendall agreement, Spearman correlation and Top-3 overlap while also producing the smallest mean rank displacement.

For instance, mean Kendall agreement increased from 0.677306 under Equal Borda to 0.746405 under primary RDRA. Similarly, mean Top-3 overlap increased from 0.720000 to 0.813333, while mean absolute candidate displacement decreased from 2.257778 to 1.773333.

The exact-winner measures are more mixed: primary RDRA does not uniquely dominate the Existing Composite on tie-aware winner survival or Top-1 set overlap. This indicates that the clearest descriptive improvement occurs at the whole-ranking and shortlist levels rather than necessarily in exact Top-1 preservation.

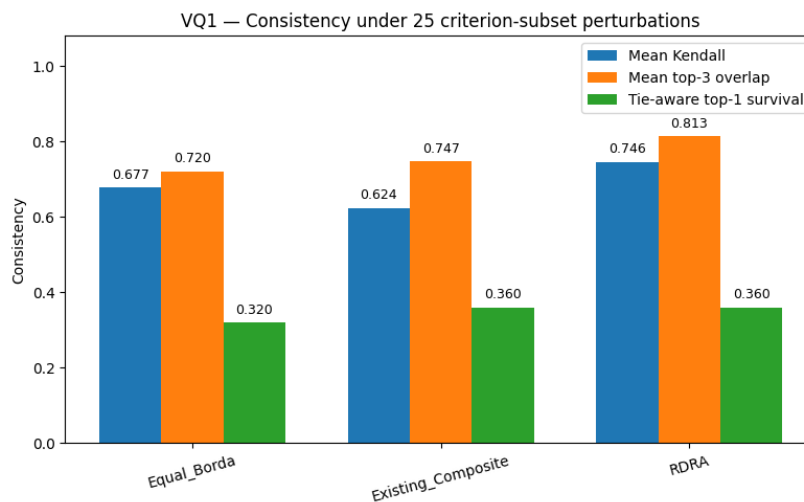


Figure 6.3 VQ1 consistency of Existing Composite, Equal Borda and primary RDRA

The figure visualises the main VQ1 consistency measures for the three aggregation methods. RDRA shows the strongest mean Kendall agreement and Top-3 preservation, while tie-aware Top-1 survival remains similar to the Existing Composite Method.

This visual comparison supports the descriptive VQ1 finding that RDRA is generally more stable under criterion uncertainty, while the following statistical-validation analysis is used to determine which improvements are statistically supported.

6.3.3 Paired Statistical Testing

The same 25 criterion perturbations were evaluated under both primary RDRA and Equal Borda. The comparison is therefore paired rather than based on independent scenario samples.

Because several VQ1 measures were tested simultaneously, Holm's sequential procedure was applied to control family-wise error [21]. Confidence intervals were retained so that the magnitude and uncertainty of each difference could be considered alongside the adjusted p -value.

	Metric	Mean improvement	95% CI	Holm-adjusted paired t-test	Effect
0	Kendall	+0.069	[0.012, 0.126]	0.039	Moderate
1	Spearman	+0.063	[-0.007, 0.134]	0.077	Small
2	Top-3 overlap	+0.093	[0.030, 0.156]	0.022	Moderate
3	Rank displacement	+0.484 better	[0.146, 0.822]	0.022	Moderate

Figure 6.4 Paired VQ1 improvements of primary RDRA over Equal Borda

The output summarises the paired statistical comparison between RDRA and Equal Borda across the 25 matched criterion-subset perturbations. RDRA achieved statistically supported improvements in Kendall agreement (+0.069), Top-3 overlap (+0.093) and rank displacement (+0.484 ranks better) after Holm correction. The Spearman improvement (+0.063) was positive but not statistically significant after correction.

This offers stronger evidence that the VQ1 consistency improvements are not only descriptive, but are statistically supported on several important measures.

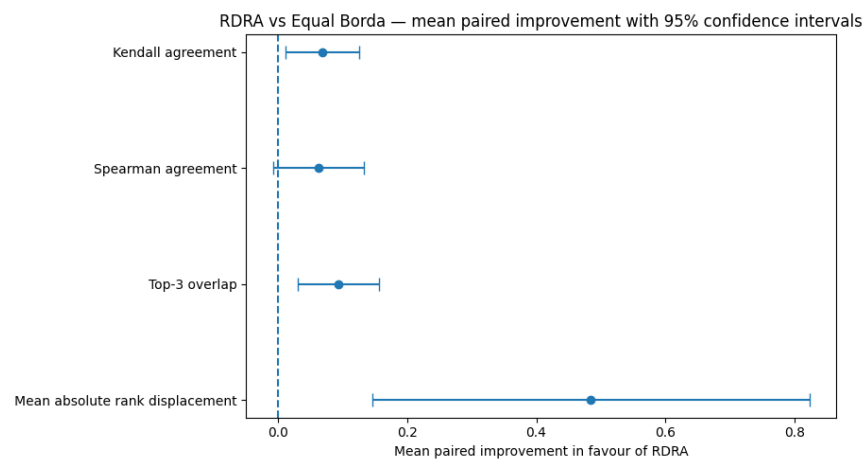


Figure 6.5 Mean paired RDRA improvement over Equal Borda with 95% confidence interval

The figure visualises the mean paired improvements and their 95% confidence intervals. The intervals for Kendall agreement, Top-3 overlap and rank displacement remain above zero, while

the Spearman interval crosses zero. This supports the bounded VQ1 result that, on some but not all evaluated measures, RDRA exhibits statistically stronger consistency than Equal Borda.

6.3.4 VQ1 Interpretation

VQ1 provides evidence that primary RDRA is less disrupted by criterion-set changes on several important measures. Therefore, the conclusion is Primary RDRA demonstrates statistically supported improvements over Equal Borda on several measures of ranking and shortlist consistency under criterion uncertainty.

The result is deliberately bounded. It does not establish significant improvement in every VQ1 measure, and exact-winner behaviour remains more mixed than whole-ranking and Top-3 consistency.

6.4 Reliability-Weight Sensitivity Testing

Because RDRA depends on empirically estimated reliability weights, an internal sensitivity test was performed to determine whether the consensus is excessively dependent on one exact estimated weight vector.

RDRA weight-estimation sensitivity								
	N_Runs	Perturbation_SD	Mean_Kendall	Median_Kendall	Mean_Top3_Overlap	Mean_Rank_Displacement	Tie_Aware_Top1_Survival	Mean_Top1_Set_Overlap
0	1000	0.05	0.964261	0.973856	1.0	0.296444	0.699	0.699

Figure 6.6 Sensitivity of RDRA consensus to reliability-weight perturbation

The primary reliability-weight vector was perturbed 1,000 times using a perturbation standard deviation of 0.05.

Each resulting vector was normalised before regenerating the candidate ranking. The perturbed rankings were then compared with the original primary RDRA consensus

The results were:

- Mean Kendall = 0.964201
- Median Kendall = 0.973856
- Mean Top-3 overlap = 1.000000
- Mean absolute rank displacement = 0.296444
- Tie-aware Top-1 survival = 0.699
- Mean Top-1 set overlap = 0.699

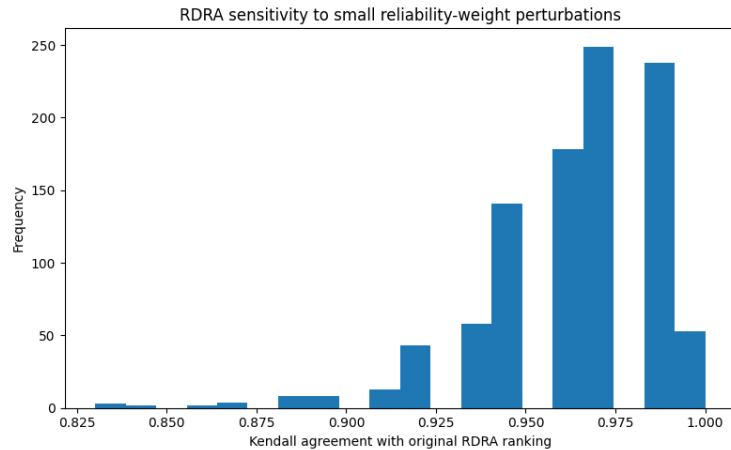


Figure 6.7 Sensitivity of the primary RDRA consensus to reliability-weight perturbation

The results indicate that the broader ranking is highly stable under the tested moderate perturbations. Mean Kendall agreement remains above 0.96 and the Top-3 shortlist is completely preserved on average.

The exact winner is more sensitive, with a survival rate of 0.699. This indicates that closely ranked candidates near the top may exchange first position even when the broader ordering remains highly stable.

This experiment is an internal sensitivity analysis, not a comparison demonstrating superiority over Equal Borda. Its purpose is to determine whether the primary RDRA consensus is excessively fragile to moderate uncertainty in the estimated reliability weights.

6.5 VQ2: Evaluation under Data-Quality Uncertainty

Where VQ1 investigates resistance to unnecessary ranking changes, VQ2 examines whether a ranking changes when the supporting evidence itself deteriorates meaningfully.

This tests a distinct property: evidence responsiveness

6.5.1 Controlled Corruption Design

The AI4I experiment contains 16 controlled data-quality perturbations drawn from four corruption families:

- Gaussian noise
- Missing Completely At Random (MCAR)
- outlier contamination
- static sensor offset

Each corruption family was evaluated at multiple severity levels.

A key methodological choice is the use of fixed clean-data normalisation bounds when evaluating corrupted evidence.

If each corrupted scenario were independently renormalised using its own minimum and maximum values, widespread deterioration could be partly hidden because degraded candidates would simply be rescaled relative to one another. Fixed clean-data bounds retain a common reference and therefore allow deterioration to remain visible as genuine change from the clean deployment evidence.

6.5.2 Evidence Change and Rank Response

For each corruption scenario, two quantities were used:

$$\Delta E = \text{deployment evidence-change magnitude}$$

and:

$$\Delta R = \text{resulting ranking-change magnitude}$$

Responsiveness is evaluated using Spearman correlation:

$$\rho = \text{Corr}_S(\Delta E, \Delta R)$$

Where ΔE = deployment evidence-change magnitude and ΔR = ranking-change magnitude.

A stronger positive association means that scenarios causing larger changes in deployment evidence tend to produce correspondingly larger changes in the ranking.

For the locked exploratory refined RDRA: $\rho_{\text{Refined RDRA}} = 0.678041$ with $p = 0.003892$.

For Equal Borda: $\rho_{\text{Borda}} = 0.347570$ with $p = 0.187141$.

The observed difference is:

$$\Delta\rho = 0.678041 - 0.347570 = 0.330471$$

	Condition	Level	Evidence_Change_Magnitude	Refined_RDRA_Rank_Displacement	Equal_Borda_Rank_Displacement
0	Gaussian Noise	0.05	0.227783	0.333333	0.888889
1	Gaussian Noise	0.10	0.301766	0.555556	1.611111
2	Gaussian Noise	0.20	0.321858	0.888889	2.333333
3	Gaussian Noise	0.30	0.305759	0.777778	2.333333
4	Missing Values MCAR	0.05	0.269785	0.333333	1.722222
5	Missing Values MCAR	0.10	0.307266	0.555556	2.055556
6	Missing Values MCAR	0.20	0.327334	1.222222	3.000000
7	Missing Values MCAR	0.30	0.322597	0.888889	3.000000
8	Outliers	0.01	0.300420	0.555556	2.388889
9	Outliers	0.03	0.313063	0.888889	3.166667
10	Outliers	0.05	0.298197	0.888889	3.277778
11	Outliers	0.10	0.286202	0.777778	3.222222
12	Static Sensor Offset	0.05	0.267268	0.333333	1.111111
13	Static Sensor Offset	0.10	0.311245	0.333333	0.777778
14	Static Sensor Offset	0.20	0.307642	0.888889	2.500000
15	Static Sensor Offset	0.30	0.296094	0.777778	2.111111

Figure 6.8 Deployment-evidence change versus ranking response across the 16 AI4I corruption scenarios

The output summarises the 16 controlled AI4I corruption scenarios and records the magnitude of deployment-evidence change together with the corresponding rank displacement for locked refined RDRA and Equal Borda. The larger positive association for the refined method suggests that ranking changes more closely track the magnitude of evidence deterioration.

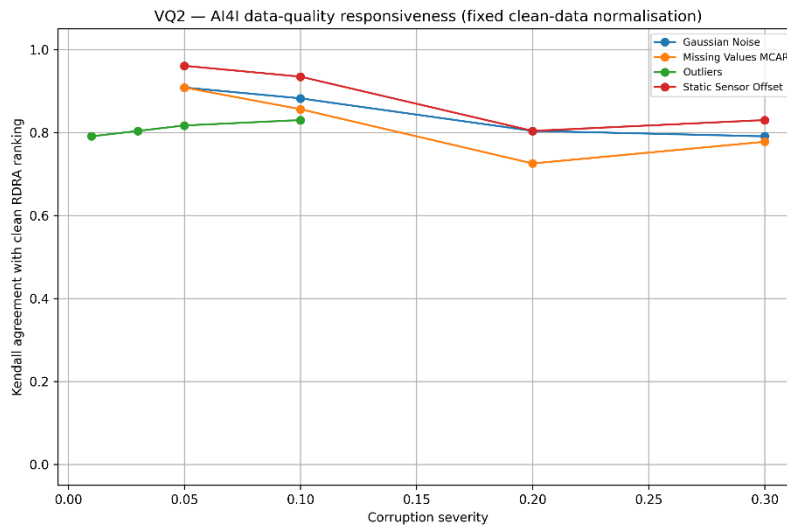


Figure 6.9 VQ2 ranking consistency under increasing AI4I corruption severity

The figure shows how agreement with the clean-data RDRA ranking changes across increasing corruption severity for the four corruption families. Ranking agreement generally remains relatively high, although the amount of change differs by corruption type and severity. This figure demonstrates that the method is not simply unstable under corrupted data, while also showing that meaningful sensor degradation can alter the resulting ranking.

6.5.3 Direct Statistical Comparison

The individual correlations alone are insufficient to establish that the two methods differ.

In particular, the inference should not be based simply on the fact that: $p_{\text{RDRA}} < 0.05$ whereas $p_{\text{Borda}} > 0.05$.

A statistically significant correlation for one method and a non-significant correlation for another does not by itself constitute a statistically significant difference between the methods.

A direct matched comparison was therefore performed using 10,000 bootstrap resamples [18].

The resulting confidence interval for $\Delta\rho = \rho_{\text{Refined RDRA}} - \rho_{\text{Borda}}$ was:

$$95\%CI = [0.046026, 0.702235].$$

Because the entire interval lies above zero, the matched-bootstrap comparison provides direct statistical support for stronger responsiveness under the tested AI4I corruption conditions.

	VQ2 statistical result	Refined RDRA	Equal Borda
0	Spearman responsiveness ρ	0.678	0.348
1	Individual two-sided p	0.0039	0.187
2	Direct difference $\Delta\rho$	+0.330	—
3	Bootstrap 95% CI for $\Delta\rho$	[0.046, 0.702]	—

Figure 6.10 Consolidated VQ2 statistical-validation result

The stronger-responsiveness conclusion is based on the direct matched-bootstrap comparison of $\Delta\rho$, rather than on contrasting the significance status of the two individual correlations. The observed difference was $\Delta\rho = +0.330$, with a 95% confidence interval of [0.046,0.702].

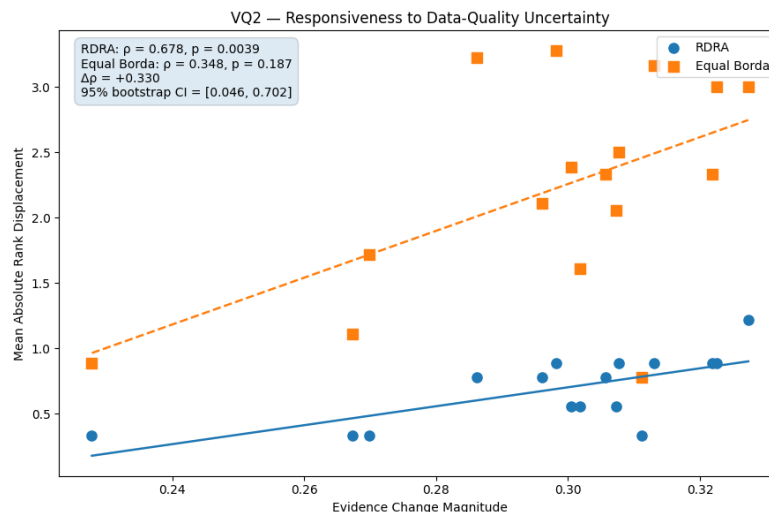


Figure 6.11 VQ2 responsiveness comparison between locked refined RDRA and Equal Borda

The figure compares the relationship between deployment-evidence change and rank displacement for locked refined RDRA and Equal Borda across the 16 corruption scenarios. Refined RDRA shows the stronger responsiveness correlation ($\rho = 0.678$) compared with Equal Borda ($\rho = 0.348$).

This provides visual evidence that refined-RDRA ranking changes are more closely aligned with changes in the underlying deployment evidence.

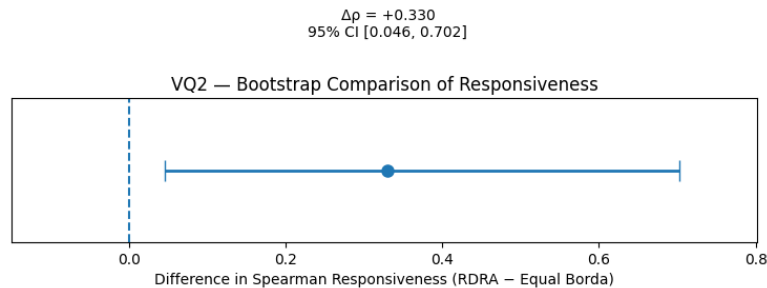


Figure 6.12 Bootstrap 95% confidence interval for the direct VQ2 responsiveness difference

The figure shows the matched-bootstrap estimate of the direct responsiveness difference between the two methods. The observed difference is $\Delta\rho = +0.330$, with a 95% confidence interval of [0.046, 0.702].

Because the interval remains above zero, this provides statistical support that the locked exploratory refined RDRA is more responsive than Equal Borda under the tested AI4I data-quality perturbations.

6.5.4 VQ2 Interpretation

The supported VQ2 conclusion is the locked exploratory refined RDRA demonstrates statistically stronger responsiveness than Equal Borda under the tested AI4I data-quality perturbations.

This conclusion applies specifically to the locked exploratory $\gamma = 3$ configuration. It should not be attributed generically to the primary $\gamma = 1$ RDRA formulation.

Combined with VQ1, the results indicate that reliability-aware aggregation can exhibit two complementary desirable properties under different uncertainty conditions: resistance to unnecessary criterion-induced disruption and responsiveness to meaningful evidence deterioration.

6.6 VQ3: Evaluation under Preference Uncertainty

VQ3 evaluates whether ranking-level behaviour translates into useful downstream model-selection choices when stakeholder deployment priorities vary.

This provides a separate decision-oriented evaluation rather than assuming that improved ranking behaviour automatically produces superior downstream utility.

6.6.1 Matched Preference-Scenario Design

A total of 5,000 stakeholder-preference scenarios were generated from a Dirichlet distribution using the fixed project random state.

Each preference vector contains non-negative weights summing to one and represents a different possible trade-off among the deployment objectives.

The same preference vectors were applied to the shortlists produced by RDRA and Equal Borda. This matched design ensures that both aggregation methods are evaluated under identical stakeholder-priority conditions.

The evaluation focuses on feasible **Top-3 shortlists** rather than only exact Top-1 selection. This reflects a realistic decision-support role in which the ranking reduces the candidate space before a final organisational or human decision.

Pareto and regret-based measures were used to evaluate the resulting shortlists.

6.6.2 Downstream Top-3 Decision Usefulness

The current VQ3 results show that RDRA and Equal Borda are broadly comparable in Top-3 downstream decision usefulness under the tested preference scenarios.

	Method	Top3_Feasible_Shortlist	Mean_Shortlist_Regret	P95_Shortlist_Regret	CVar95_Shortlist_Regret	Worst_Case_Shortlist_Regret	Zero_Regret_Frequency
0	Equal_Borda	XGBoost class_weight isotonic XGBoost ...	0.002069	0.016802	0.038217	0.079113	0.9316
1	RDRA	XGBoost class_weight isotonic XGBoost ...	0.002069	0.016802	0.038217	0.079113	0.9316
2	Existing_Composite	XGBoost class_weight platt XGBoost cla...	0.003082	0.019930	0.057028	0.139577	0.9260

Figure 6.13 VQ3 Top-3 downstream regret across 5,000 matched stakeholder-preference scenarios

The output summarises the Top-3 shortlist and regret results for RDRA, Equal Borda and the Existing Composite Method across 5,000 matched preference scenarios. RDRA and Equal Borda produce very similar shortlist-regret results, while the Existing Composite Method shows slightly higher regret on the reported measures.

This provides evidence that the VQ3 downstream decision-usefulness analysis was completed and shows that RDRA does not currently demonstrate an additional Top-3 decision-usefulness

advantage over Equal Borda. Therefore, the result is used to define the current contribution boundary rather than to claim superiority.

6.6.3 Joint Preference and Corruption Evidence

The implementation also evaluates candidate behaviour when stakeholder-preference uncertainty and corrupted evidence occur simultaneously

	Variant	Mean_Joint_Regret	P95_Joint_Regret	Worst_Case_Joint_Regret	Zero_Regret_Frequency	Joint_Robust_Regret_Rank
3	XGBoost class_weight platt	0.123966	0.560538	0.969285	0.089765	1
2	XGBoost class_weight isotonic	0.084179	0.534891	0.971801	0.686618	2
0	Random Forest class_weight isotonic	0.533107	0.981915	0.998236	0.043500	3
1	Random Forest class_weight platt	0.530673	1.000000	1.000000	0.180118	4

Figure 6.14 VQ3 joint preference-plus-corruption regret analysis

The joint analysis indicates that several XGBoost configurations achieve stronger candidate-level robust-regret positions than the evaluated Random Forest configurations under the combined uncertainty conditions.

However, this result is primarily candidate-level robustness evidence. It should not be interpreted as direct RDRA-versus-Equal-Borda evidence unless the aggregation methods themselves are compared under the same matched joint conditions.

Hence, its role is to identify model configurations that remain comparatively defensible under simultaneous preference and data-quality uncertainty.

6.6.4 VQ3 Interpretation

VQ3 establishes an important boundary on the current contribution.

The positive ranking-level results observed elsewhere do not automatically translate into improved Top-3 downstream decision usefulness under varying stakeholder preferences. This confirms the value of evaluating consistency, responsiveness and downstream usefulness as separate properties rather than collapsing them into a single claim of method superiority.

Accordingly, the current evidence supports

- VQ1: several consistency advantages
- VQ2: stronger responsiveness for the locked refinement
- VQ3: no additional Top-3 advantage currently demonstrated

The neutral VQ3 result is therefore part of the scientific contribution boundary rather than a result to be excluded from the evaluation.

6.7 Complementary C-MAPSS Cross-Task Evaluation

C-MAPSS FD001 provides a complementary predictive-maintenance context substantially different from the primary AI4I classification experiment [7], [21], [22].

Whereas AI4I addresses binary machine-failure classification, C-MAPSS addresses sequential Remaining Useful Life prediction.

The evaluated C-MAPSS candidate models include:

- Ridge
- Random Forest
- XGBoost
- LSTM
- GRU

The current implementation examines three deployment dimensions:

- Predictive Quality
- Robustness
- Computational Efficiency

Controlled temporal data degradation includes conditions such as Gaussian noise, gradual offset drift, missing blocks, outliers and sensor freezing.

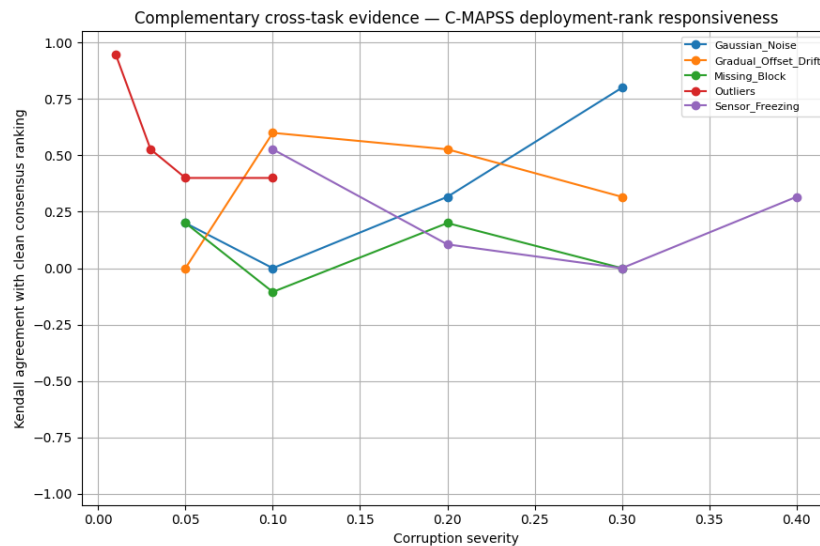


Figure 6.15 C-MAPSS complementary cross-task deployment-rank responsiveness

The figure shows Kendall agreement with the clean-data consensus ranking across different corruption types and severity levels. The ranking response varies across Gaussian noise, gradual offset drift, missing blocks, outliers and sensor freezing rather than following one identical pattern.

This provides complementary evidence that deployment rankings are sensitive to data-quality changes in a different predictive-maintenance setting involving RUL prediction, while not claiming that C-MAPSS constitutes a second full RDRA validation.

The C-MAPSS branch provides additional breadth because the ranking problem is examined using a different target variable, data structure and model family.

However, the current C-MAPSS pipeline does not contain the same repeated five-dimension evidence structure required to estimate a full independent C-MAPSS reliability-weight vector.

6.8 Overall Evaluation against the Research Challenge and Requirements

The complete evaluation provides evidence at three levels: **functionality**, **ranking behaviour** and **downstream usefulness**.

Table 6.2 Overall evaluation of the implemented RDRA solution

Evaluation objective	Scientific evidence	Assessment
Does the artefact provide the required functionality?	Five-dimension scoring, criterion ranking, LOO reliability estimation, Borda aggregation and persistent outputs completed	Achieved
Is the resulting consensus aligned with straightforward deployment evidence?	9 comparable dominance pairs, 0 violations	Supported
Does primary RDRA improve consistency under criterion uncertainty?	Significant improvements in Kendall, Top-3 overlap and rank displacement after Holm correction	Supported on several measures
Is primary RDRA robust to moderate uncertainty in its estimated weights?	Mean Kendall 0.964201 and Top-3 overlap 1.0 across 1,000 perturbations	Supported
Does reliability-aware ranking respond meaningfully to data-quality deterioration?	Refined RDRA $\rho = 0.678041$; direct $\Delta\rho$ bootstrap CI entirely above zero	Supported for locked exploratory refined RDRA
Does RDRA improve Top-3 downstream usefulness under changing stakeholder preferences?	RDRA and Equal Borda broadly comparable under 5,000 matched scenarios	No additional advantage currently demonstrated

Is there supporting evidence outside AI4I classification?	C-MAPSS sequential RUL experiment	Complementary evidence within defined scope
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The overall results show that RDRA addresses the research challenge in a specific manner. Primary RDRA provides statistically supported gains on several measures of consensus consistency under criterion uncertainty, while the weight-sensitivity experiment indicates that its broader ranking structure is not excessively dependent on minor variations in the empirical weight estimates.

The locked exploratory refinement additionally provides statistically supported evidence of stronger responsiveness to data-quality deterioration than Equal Borda under the tested AI4I corruption scenarios. The direct matched-bootstrap comparison is particularly important because it tests the difference between the methods rather than relying on separate significance decisions.

At the same time, VQ3 does not demonstrate an additional Top-3 downstream usefulness advantage under the tested preference scenarios. This result establishes an important empirical boundary: improved ranking consistency and responsiveness do not necessarily guarantee superior downstream utility for every decision setting. This bounded overall interpretation is consistent with the current Research Implementation evidence

This interpretation is consistent with the purpose of RDRA. The proposed method is intended to make criterion influence reflect the empirical reproducibility of the supporting evidence, rather than guarantee one universally optimal candidate under all operational conditions. Therefore, the evidence supports the more defensible conclusion, which is Reliability-aware aggregation provides measurable value under forms of deployment uncertainty.

6.9 Section Summary

The evaluation confirms that the implemented RDRA artefact satisfies the defined functional requirements and remains aligned with the directly comparable deployment evidence. Under VQ1, primary RDRA demonstrates statistically supported improvements over Equal Borda on several measures of ranking and shortlist consistency, while the reliability-weight sensitivity analysis indicates strong stability of the broader consensus under moderate weight perturbations.

Under VQ2, the locked exploratory refined RDRA demonstrates statistically stronger responsiveness than Equal Borda under the tested AI4I data-quality perturbations. In contrast, VQ3 does not currently demonstrate an additional Top-3 downstream decision-usefulness advantage under the matched stakeholder-preference scenarios. C-MAPSS provides complementary cross-task responsiveness evidence within the defined scope.

Therefore, these findings establish a **bounded empirical contribution**: RDRA provides measurable benefits for particular ranking properties and uncertainty conditions, but the current evidence does not support universal superiority across all downstream decision measures.

7. Conclusion

This research implementation developed and validated **Reliability-Aware Deployment Rank Aggregation (RDRA)** for deployment-oriented predictive-maintenance model selection. The method addresses the problem of combining conflicting deployment criteria when the empirical evidence supporting those criterion-specific rankings differs in reproducibility.

The proposed concept was implemented as a functional Python/Google Colab computational research artefact. The completed pipeline constructs the five deployment dimensions, generates tie-aware criterion rankings, estimates Leave-One-Repetition-Out ranking reproducibility, converts empirical stability into reliability weights, and produces Equal Borda and primary RDRA consensus rankings through a shared aggregation framework.

The scientific evaluation shows that the contribution is **useful but bounded**. Under VQ1, primary RDRA demonstrated statistically supported improvements over Equal Borda on several measures of ranking and shortlist consistency under criterion uncertainty. The reliability-weight sensitivity analysis also showed that the broader RDRA consensus remains highly stable under moderate perturbations of the estimated weights. Under VQ2, the locked exploratory refined RDRA demonstrated statistically stronger responsiveness than Equal Borda under the tested AI4I data-quality perturbations. In contrast, VQ3 did not demonstrate an additional Top-3 downstream decision-usefulness advantage under the tested stakeholder-preference scenarios. C-MAPSS FD001 further provided complementary cross-task evidence in a different Remaining Useful Life prediction setting.

Overall, the implementation demonstrates that empirical ranking reproducibility can be incorporated into deployment-oriented rank aggregation in a transparent and testable manner. The current evidence supports reliability-aware aggregation provides measurable value under particular forms of deployment uncertainty. Therefore, the project has progressed from conceptual design to a broadly completed computational artefact with traceable outputs and scientific validation evidence, providing a sound basis for the subsequent Research Analysis stage.

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